TradeLENS: CNN-Based Trading Strategy with Integrated Technical and Macroeconomic Representations

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Introduction

Financial markets are driven by price, indicators, and macroeconomics.

Deep learning, especially CNNs, enables image-based prediction.

- However, existing CNN models are usually trained on technical indicators only, and they rarely encode macroeconomic context in the same input representation.
- Motivation: Close the gap by integrating macroeconomic context.

Numerical Models

- Use technical indicators as numeric inputs.
- Example:
  - Chakole et al. (2021) — Deep Trend Following models using CNNs show improved performance over buy-and-hold
  - Saud et al. (2024) — LSTM and GRU models using MACD, DMI, and KST demonstrate that MACD-driven signals are the most reliable in their evaluation

Image-Based Trading Approaches

- Transform time-series into visual representations (e.g., candlesticks, bar charts).
- Cohen et al. (2019) — Framed trading as image classification; inspired by trader chart analysis.
- Liu et al. (2023) — Models using candlestick images outperform numeric ones.

CNN-Based Technical Indicator Clustering

- Sezer & Özbayoglu (2018) — Early works transformed technical indicators into 15×15 grayscale matrices for CNN models
- Maratkhan et al. (2021) — Extended with CNN-TIC, EO-CNN-TIC, ResNet-TA.
  - Achieved Recall up to 0.94 (Buy) / 0.96 (Sell)
  - Annual return up to 36.7% on DOW-30

Multimodal Fusion Models

- Combine different data types such as prices, indicators, and sentiment or policy text (Wang, Mo, Modak)
- Studies show that macroeconomic indices such as MAI, FII, and EPU improve predictive performance and explain market movements (Ma, Chauhan)

RESEARCH GAP

1. Prior CNN studies focus on technical indicators only.

2. No established framework for macro-informed CNN image inputs

3. This leaves a gap for a macro informed CNN image representation that can be tested both as a classifier and as a trading system.

GENERAL OBJECTIVE

Develop and evaluate an enhanced CNN-based trading model that integrates both technical and macroeconomic indicators to assess their effect on financial market trend prediction and trading performance.

SPECIFIC OBJECTIVES

To enhance the CNN-TIC framework by expanding its technical indicator image matrix to include macroeconomic variables, forming a unified visual representation.

To compare model performance using metrics such as accuracy, F1-score, and annualized return, to determine whether the inclusion of macroeconomic indicators leads to significant improvements.

To evaluate performance on different stock categories in PSE market and cryptocurrency.

To evaluate feature importance and model behavior, interpret how macroeconomic information contributes to CNN decision-making in the context of financial trend prediction.

SCOPE AND LIMITATION

SCOPE

Philippine stocks (blue-chip + penny) and cryptocurrency.

Fixed-size indicator image dataset (15×20 for technical, 20×20 for technical+macro)

Comparison: CNN–TIC vs. CNN–MIC.

LIMITATION

Focused on Philippine stocks only.

Includes fixed commission per trade

Ignores slippage

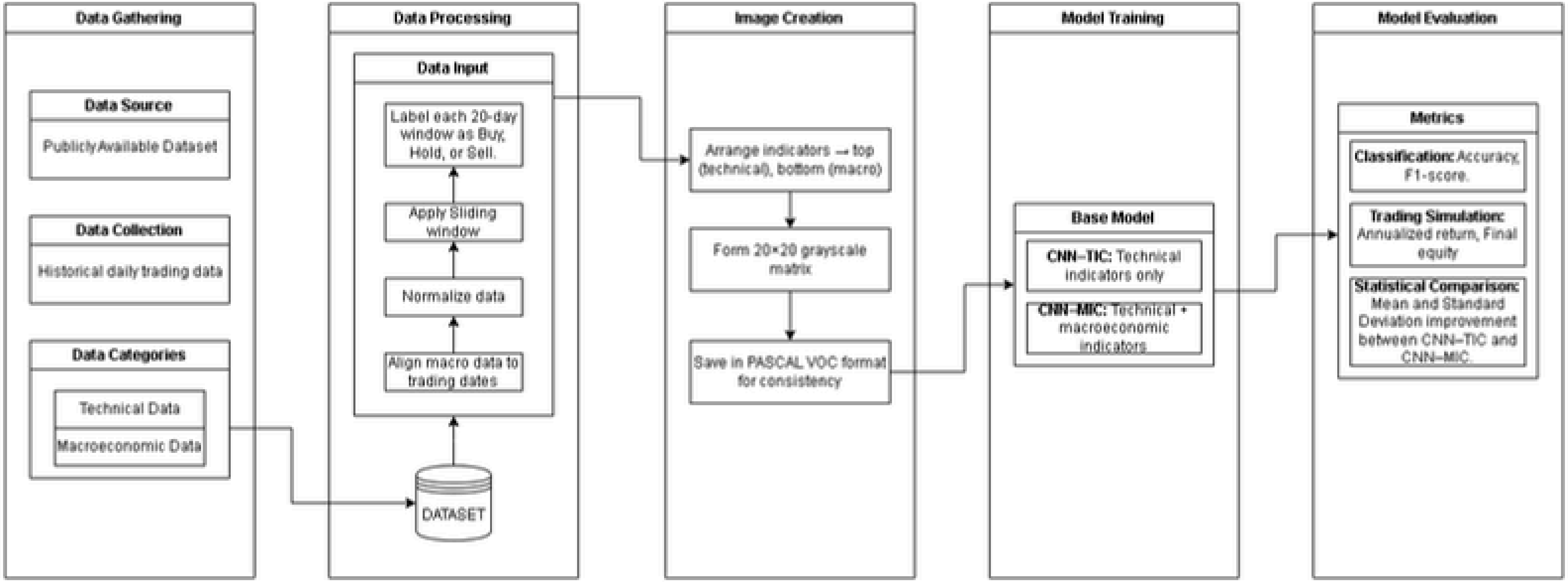
SIGNIFICANCE OF THE STUDY

Enhances robustness and accuracy of CNN-based trading models.

Supports data-driven decision-making in varying market conditions.

Extends computer vision applications in finance.

Proposed Approach



Methodology

Data Set → Data Preprocessing → Image Creation → Model Training

1

Historical data

2

Technical indicators

3

Macroeconomic Indicators

Methodology

Data Set → **Data Preprocessing** → Image Creation → Model Training

Align macro data to trading dates → Normalize values → Apply sliding window → Label each 11-day window as Buy, Hold, or Sell

Methodology

Algorithm 1 Turning point label generation (Buy, Hold, Sell)

```
procedure LABELGEN_TURNINGPOINT
    N = length(close_prices)                ▷ close_prices = time series of daily close prices
    R = 5                                   ▷ 11 day window, 5 days before and after
    for t = 0 to N − 1 do
        start = max(0, t − R)
        end = min(N − 1, t + R)
        window = close_prices[start..end]
        center_price = close_prices[t]
        max_price = max(window)
        min_price = min(window)
        max_count = number of positions in window equal to max_price
        min_count = number of positions in window equal to min_price
        if center_price = max_price and max_count = 1 then
            label_text[t] = "Sell"
            label_code[t] = −1
        else if center_price = min_price and min_count = 1 then
            label_text[t] = "Buy"
            label_code[t] = +1
        else
            label_text[t] = "Hold"
            label_code[t] = 0
        end if
    end for
    save table with columns: dates, close_prices, label_text, label_code
    return label_code
end procedure
```

LABEL GENERATION

- Identifies Buy, Hold, and Sell points
- Uses an 11-day symmetric window
- Marks local maxima as Sell and local minima as Buy
- Defaults to Hold when not a unique extreme
- Applied to every trading day

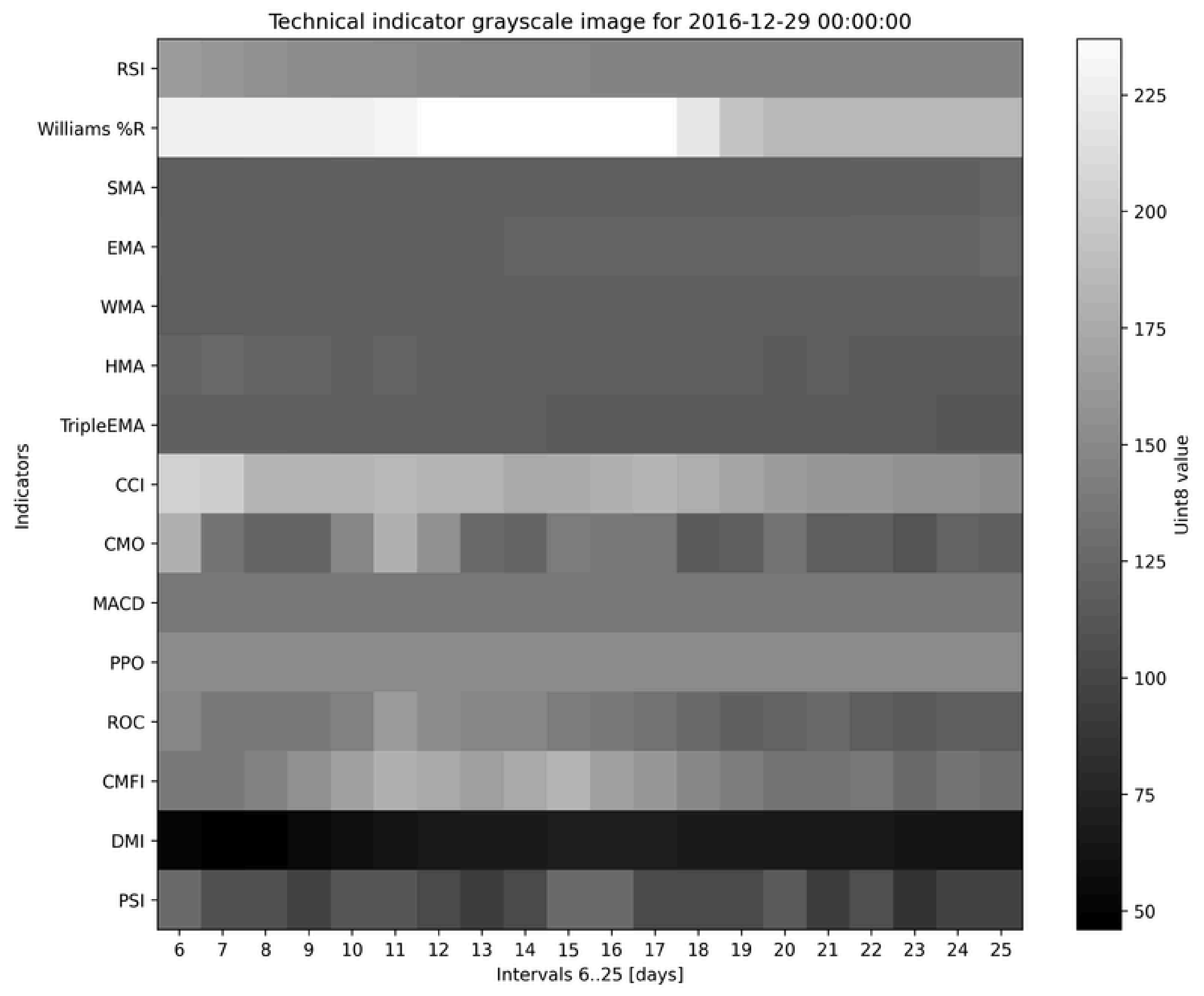
Methodology

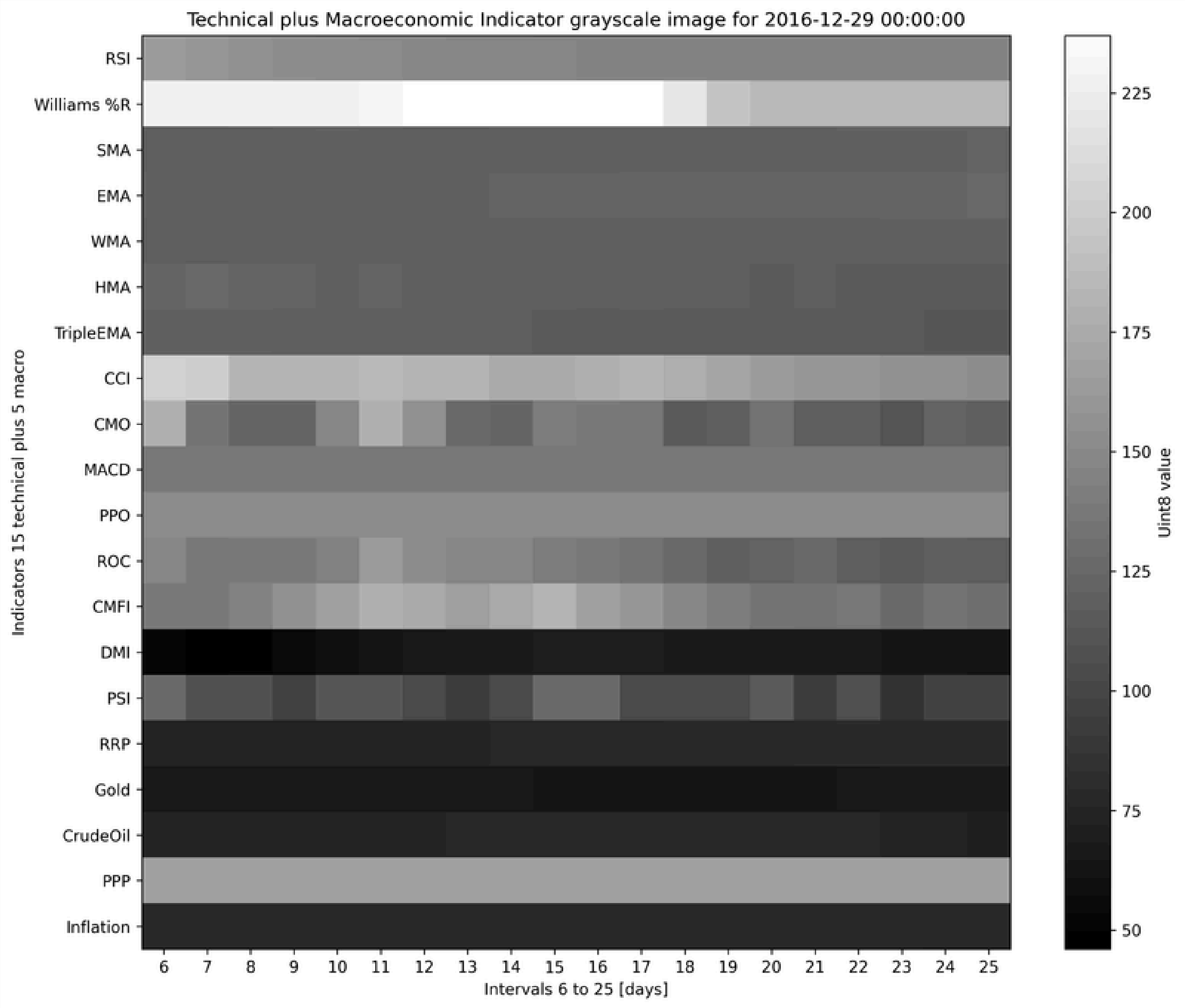
Data Set

Data Preprocessing

Image Creation

Model Training





Methodology

Data Set → Data Preprocessing → Image Creation → **Model Training**

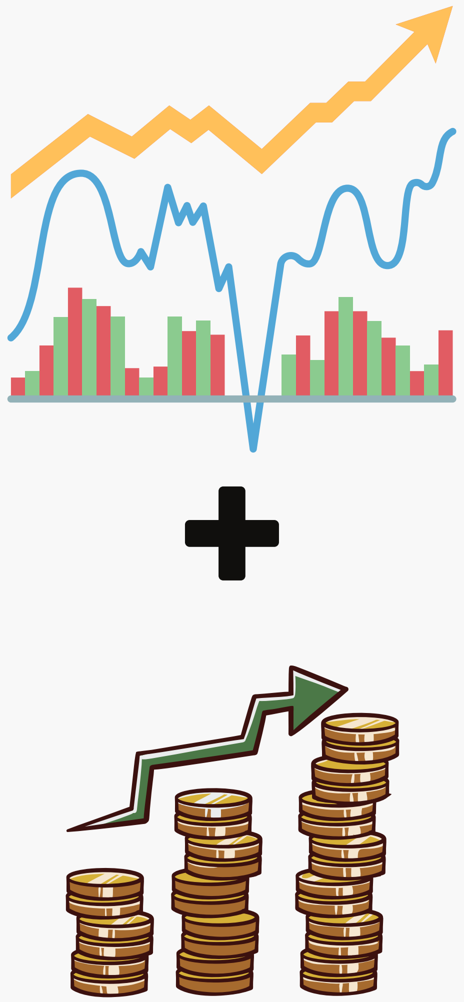
CNN–TIC: TECHNICAL INDICATORS ONLY

Historical data | Indicator | Image | CNN | Prediction

Methodology

Data Set → Data Preprocessing → Image Creation → **Model Training**

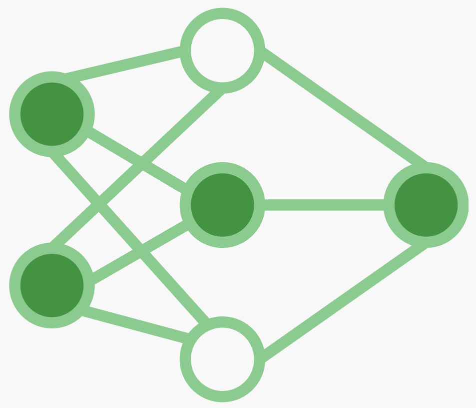
PROPOSED MODEL CNN-MIC: TECHNICAL + MACROECONOMIC INDICATORS



Historical data

Indicator

Image



CNN

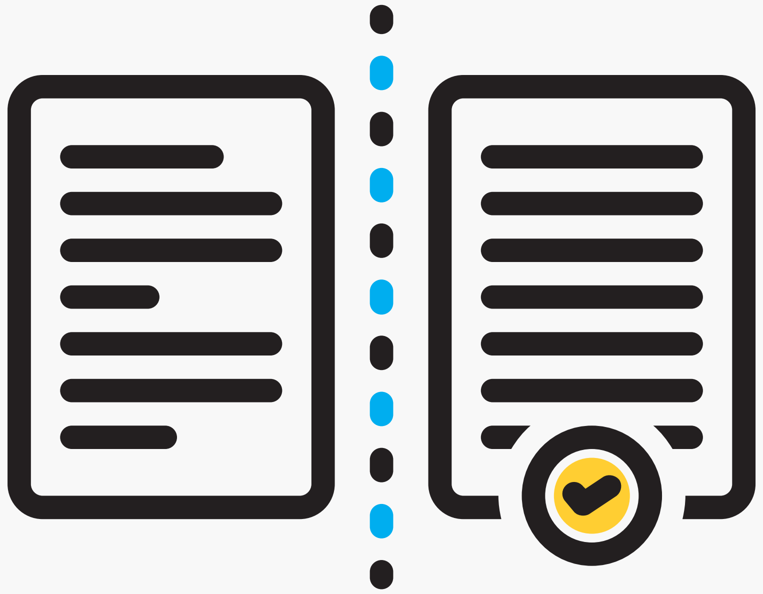
Prediction

Methodology

et → Data Preprocessing → Image Creation → Model Training → **Model Evaluation**

Classification

Trading Simulation



Statistical Comparison

Methodology

Algorithm 2 FinancialEvaluation

```
procedure FINANCIALEVALUATION
    balance = 10000 PHP
    bought_amount = 0
    commission = 50 PHP
    take_profit = 10%
    stop_loss = 1%
    for all (price, predicted_action) in testDataset do
        if bought_amount ≠ 0 then
            if price > take_profit_price then
                balance = bought_amount * take_profit_price − commission
                bought_amount = 0
            end if
            if price < stop_loss_price then
                balance = bought_amount * stop_loss_price − commission
                bought_amount = 0
            end if
        end if
        if predicted_action == BUY and balance ≠ 0 then
            bought_amount = (balance − commission)/price
            balance = 0
            take_profit_price = price * (1.0 + take_profit/100)
            stop_loss_price = price * (1.0 − stop_loss/100)
        end if
        if predicted_action == SELL and bought_amount ≠ 0 then
            balance = bought_amount * price − commission
            bought_amount = 0
        end if
    end for
    return balance
end procedure
```

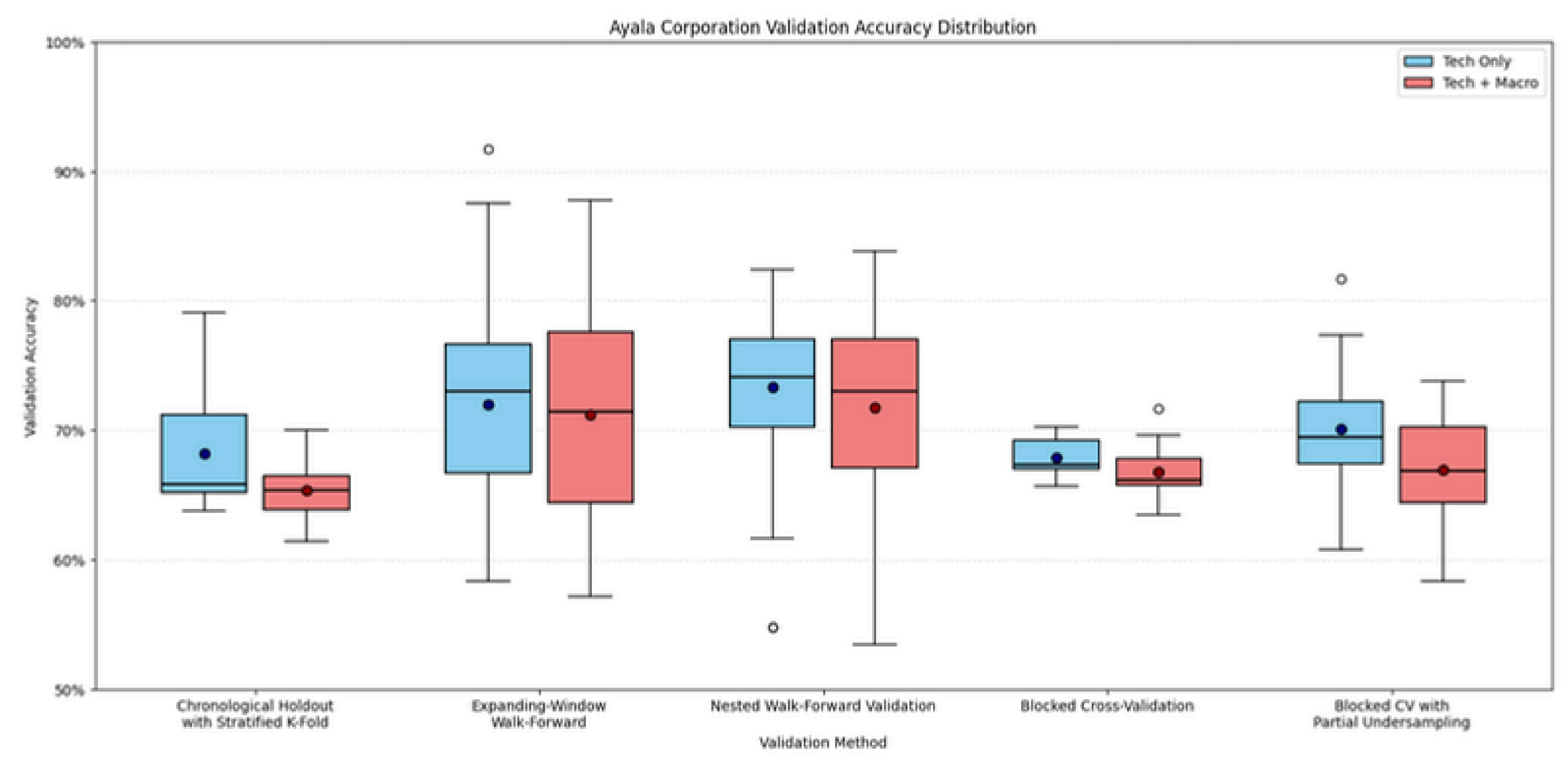
FINANCIAL EVALUATION

- initial capital 10 000 PHP
- 50 PHP commission per transaction
- take profit 10 percent, stop loss 1 percent

Blue-Chip Stocks Training Results

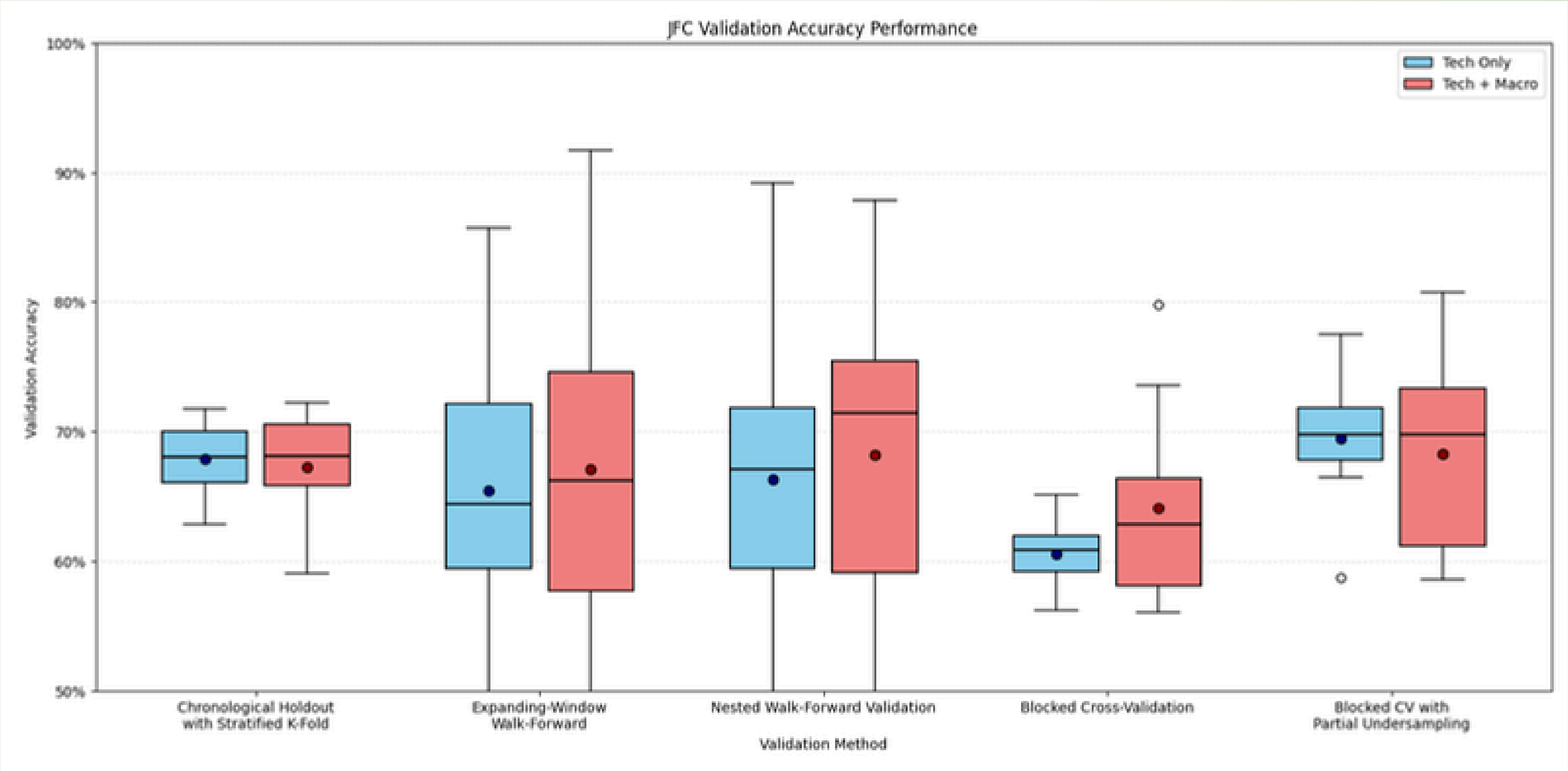
Ayala Corporation

VALIDATION ACCURACY

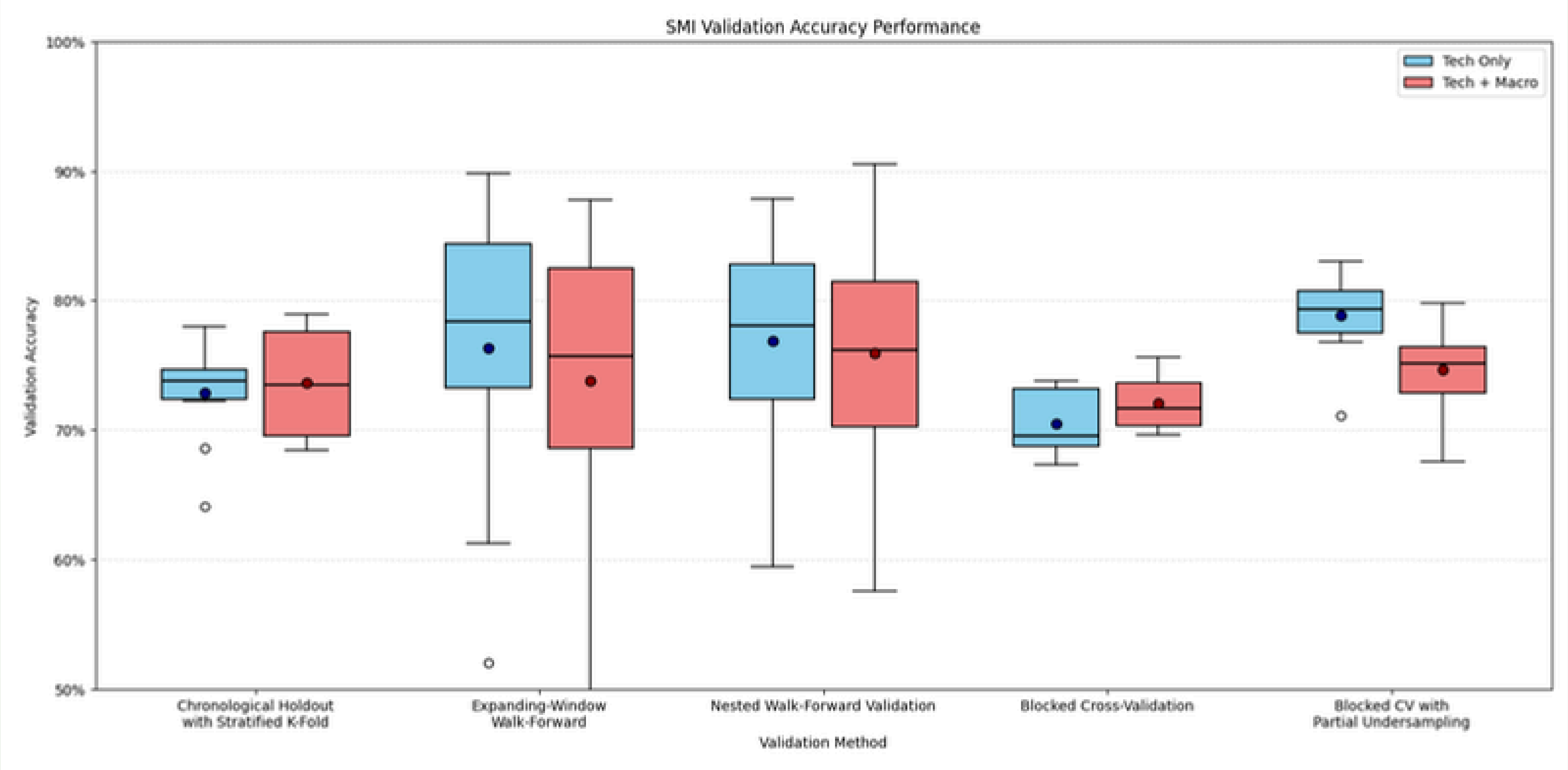


Ayala Corporation

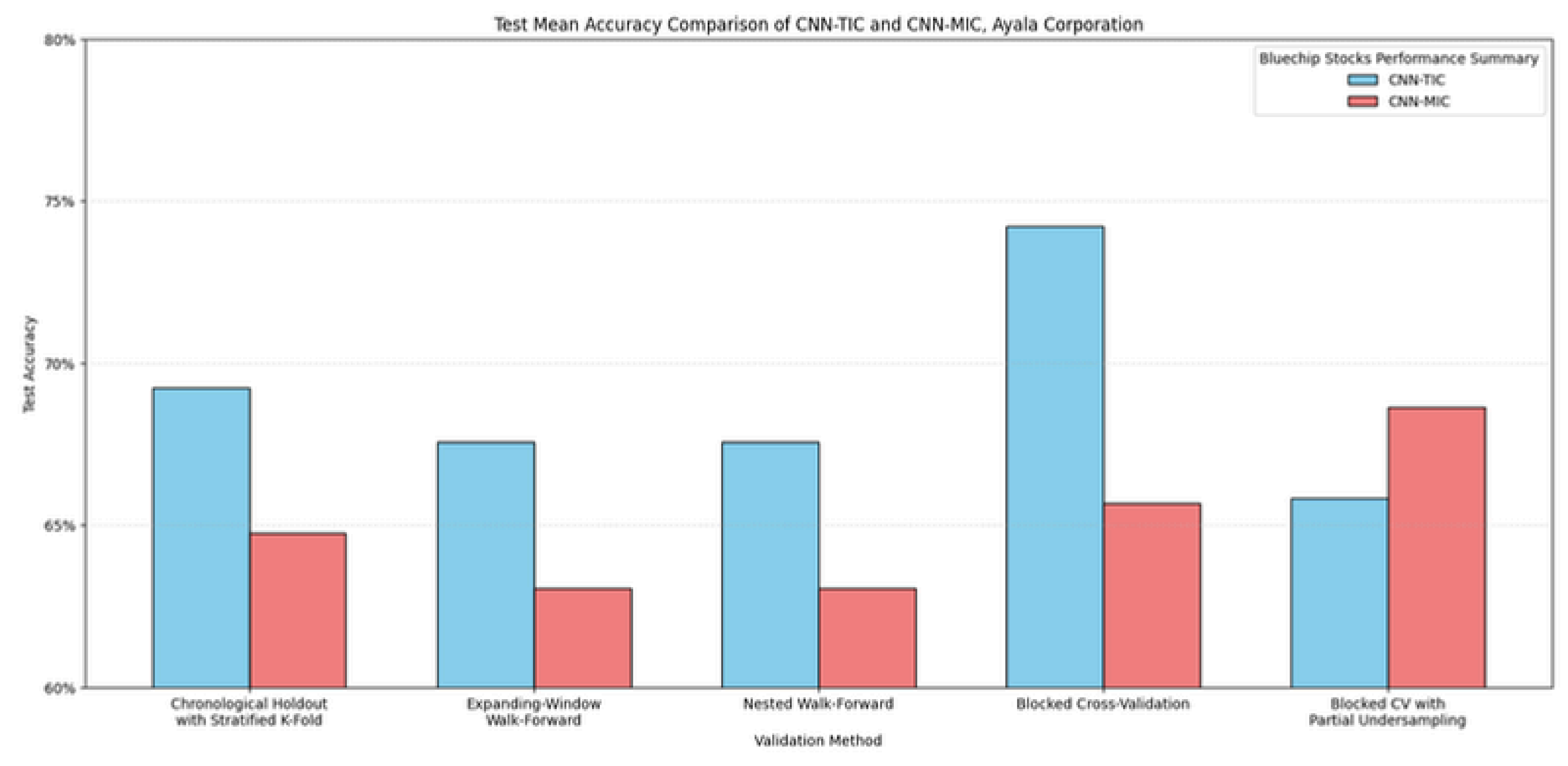
Jollibee Food Corporation



SM Investments Corporation

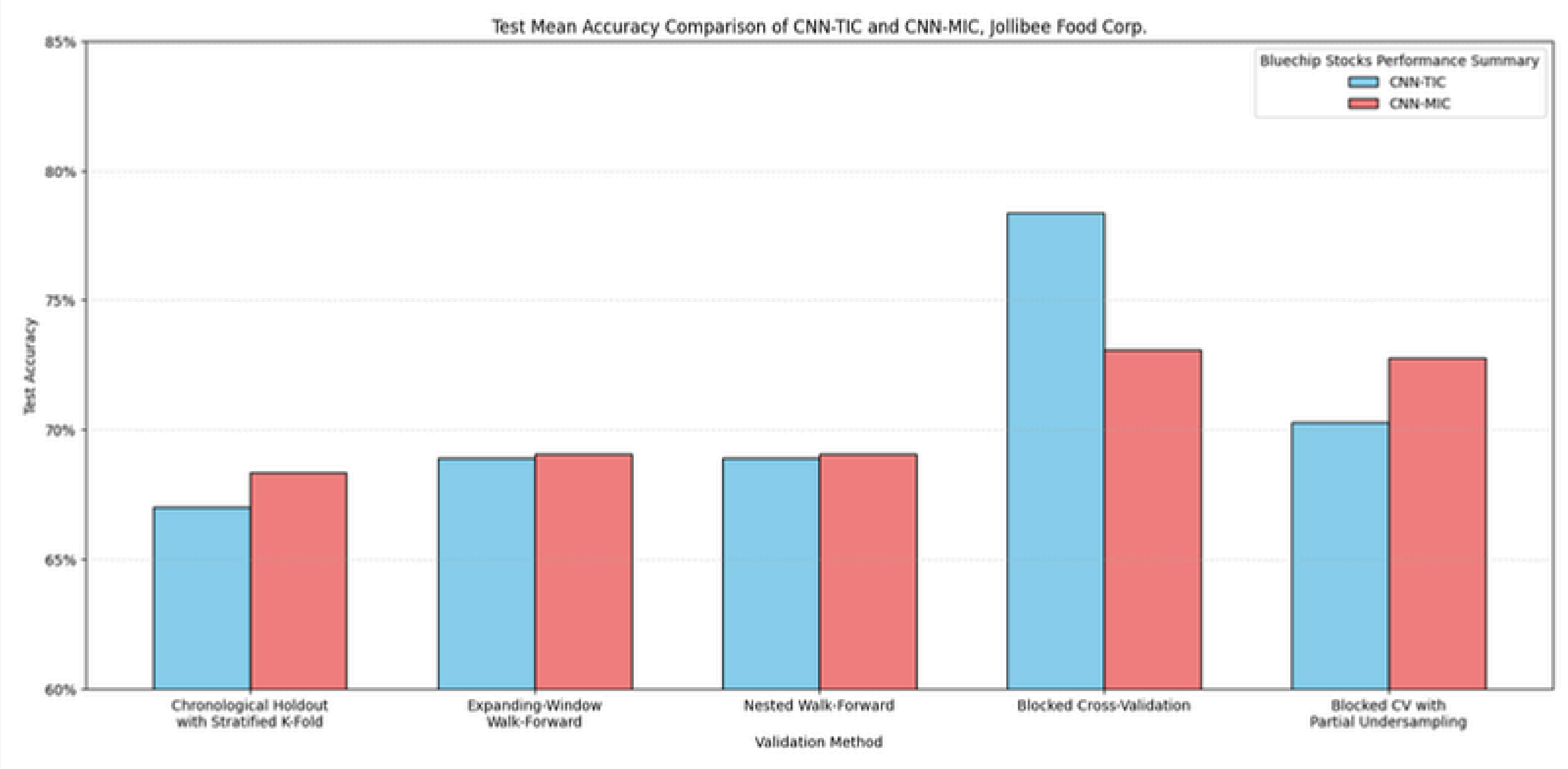


TEST ACCURACY

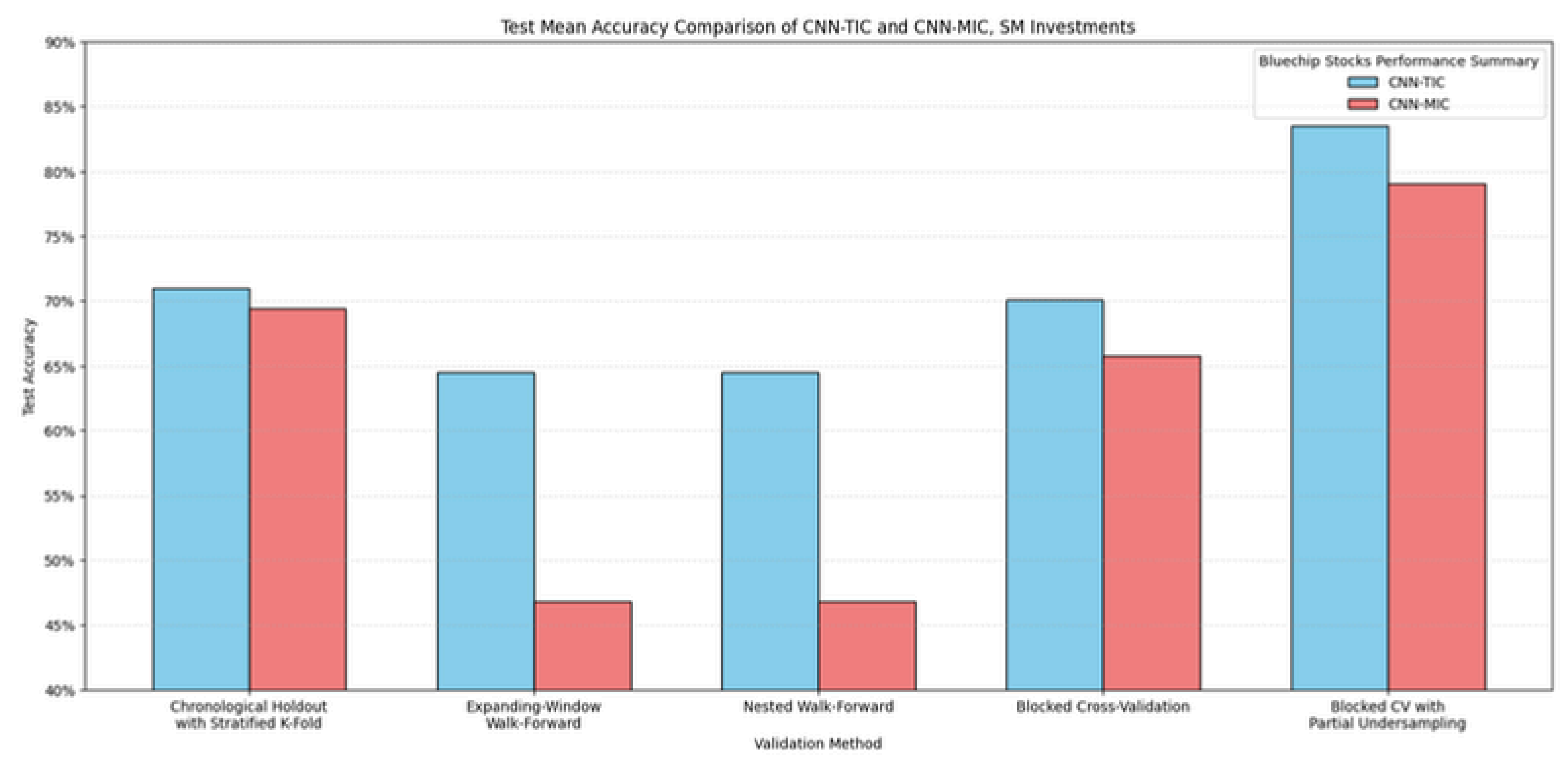


Ayala Corporation

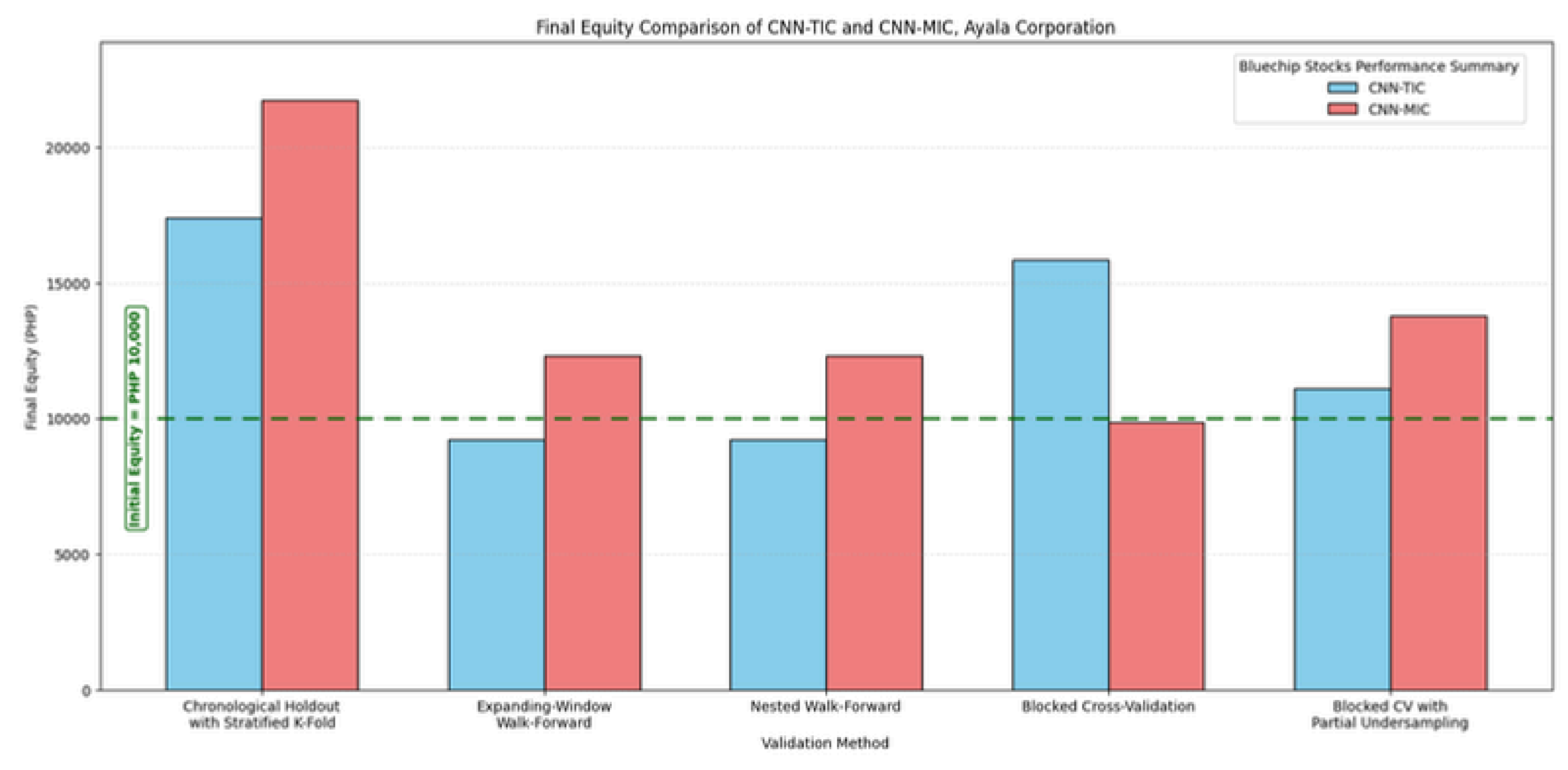
Jollibee Food Corporation



SM Investments Corporation

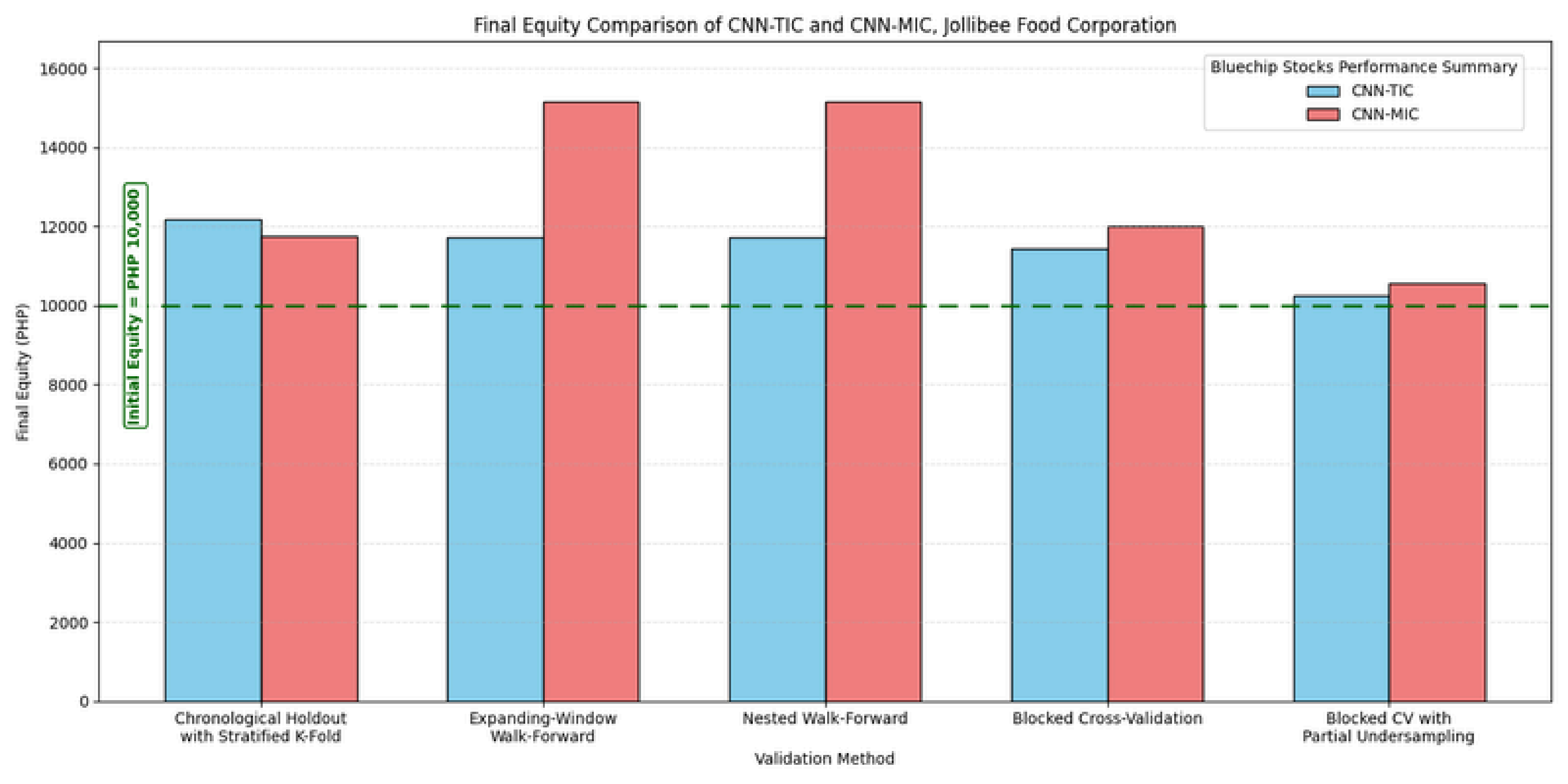


FINANCIAL PERFORMANCE

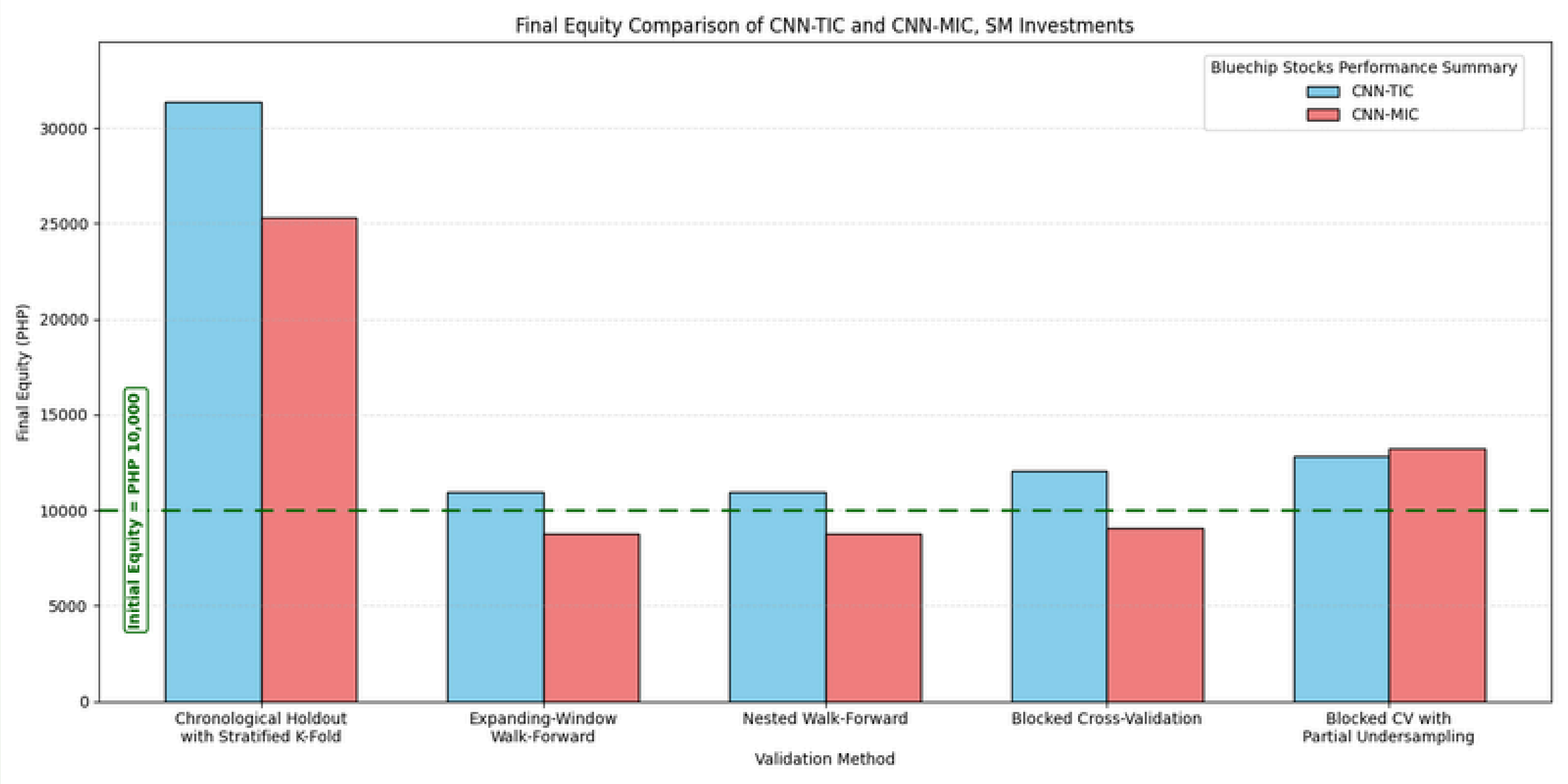


Ayala Corporation

Jollibee Food Corporation



SM Investments Corporation



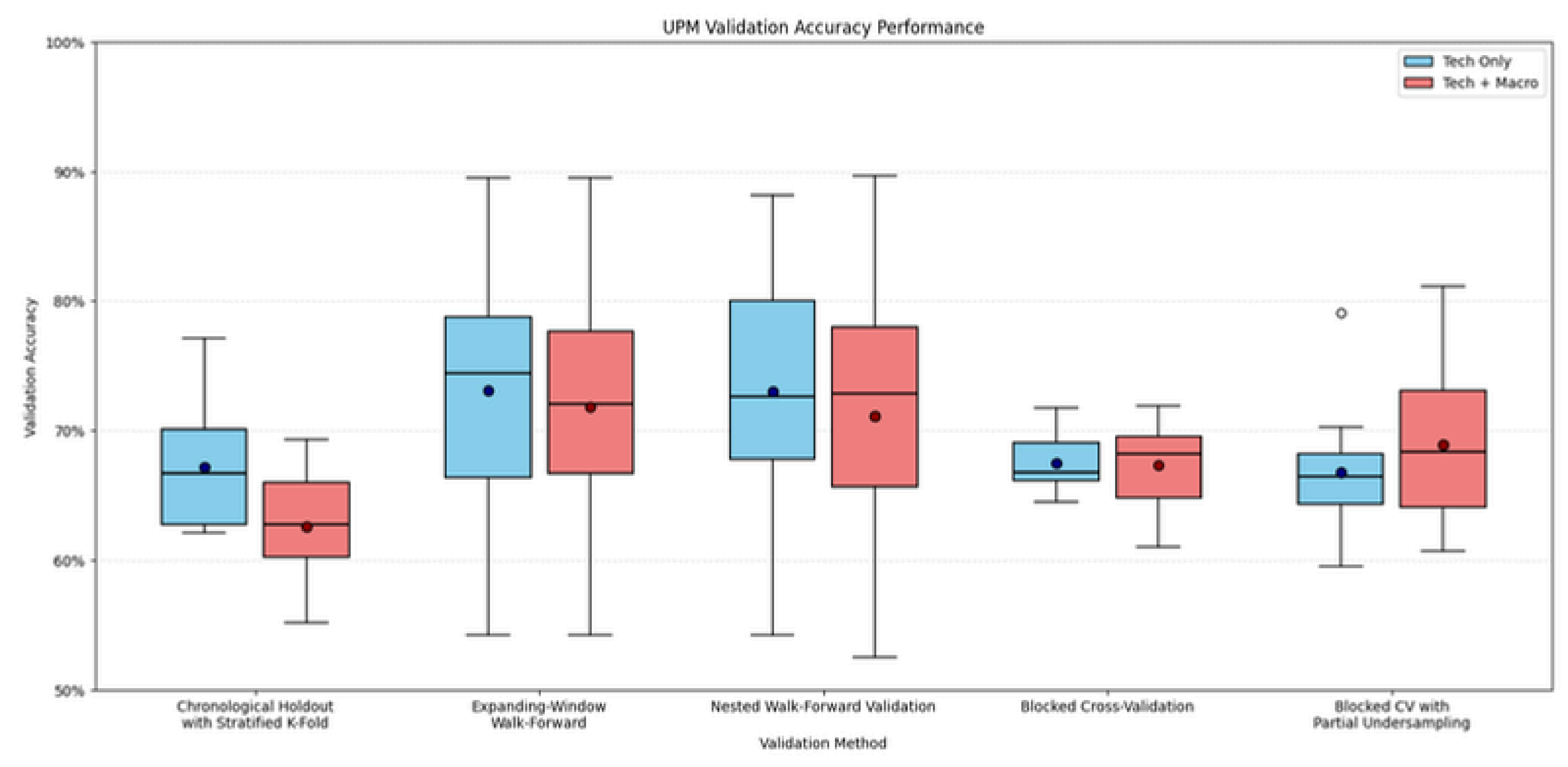
Penny Stocks Training Results

UNITED PARAGON MINING CORPORATION

PH RESORTS GROUP

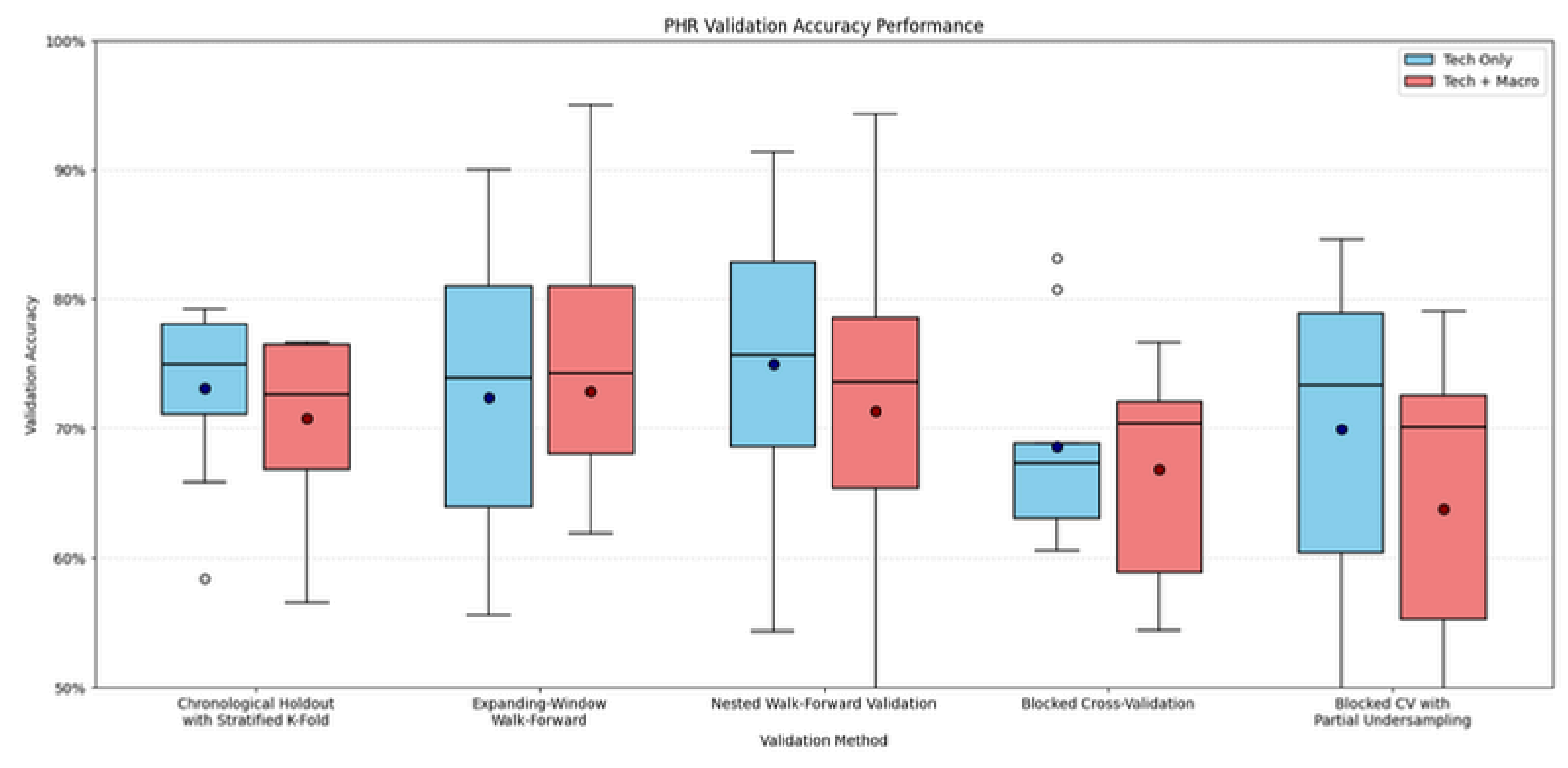
ARANETA PROPERTIES

VALIDATION ACCURACY

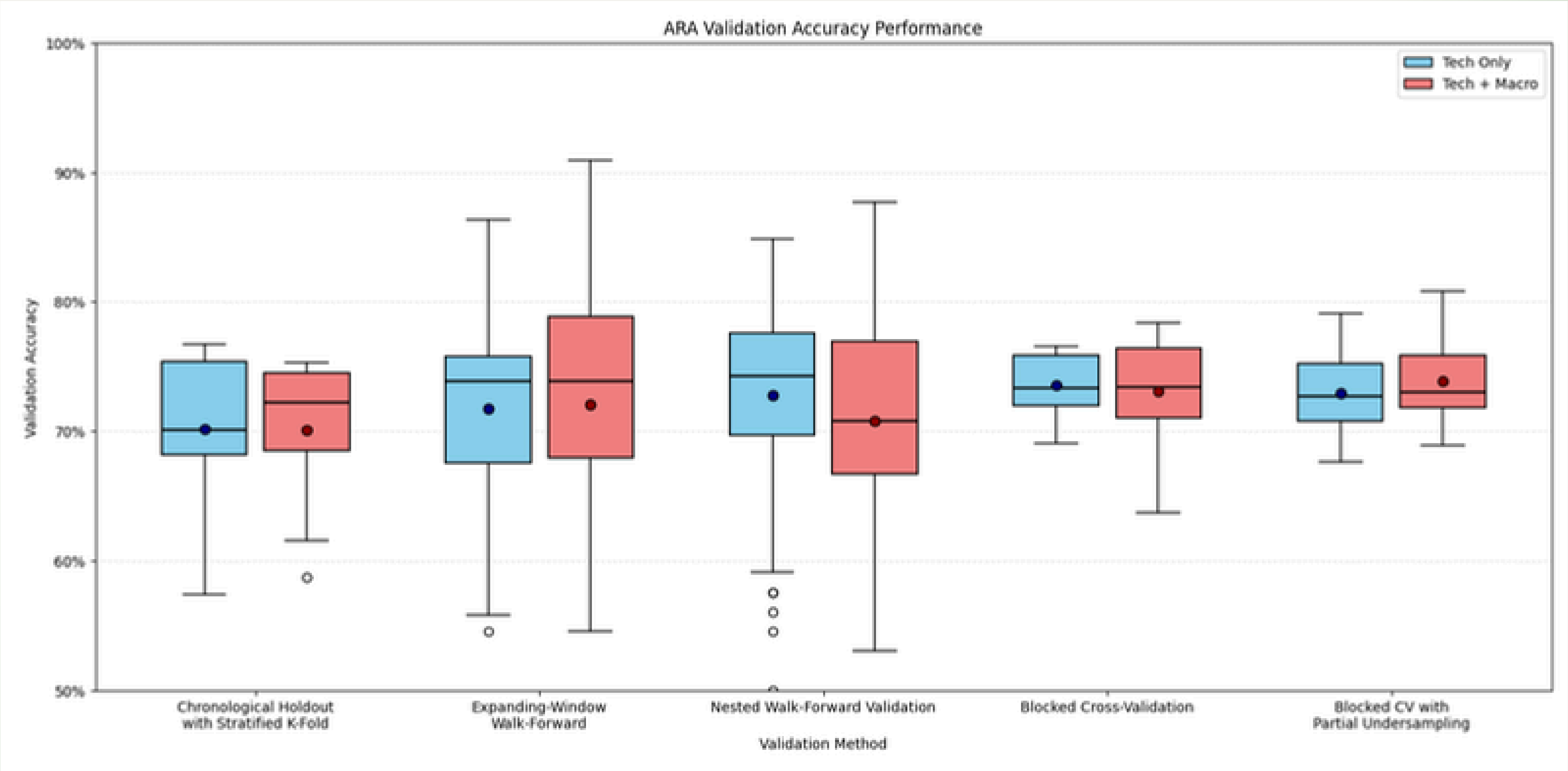


United Paragon Mining Corporation

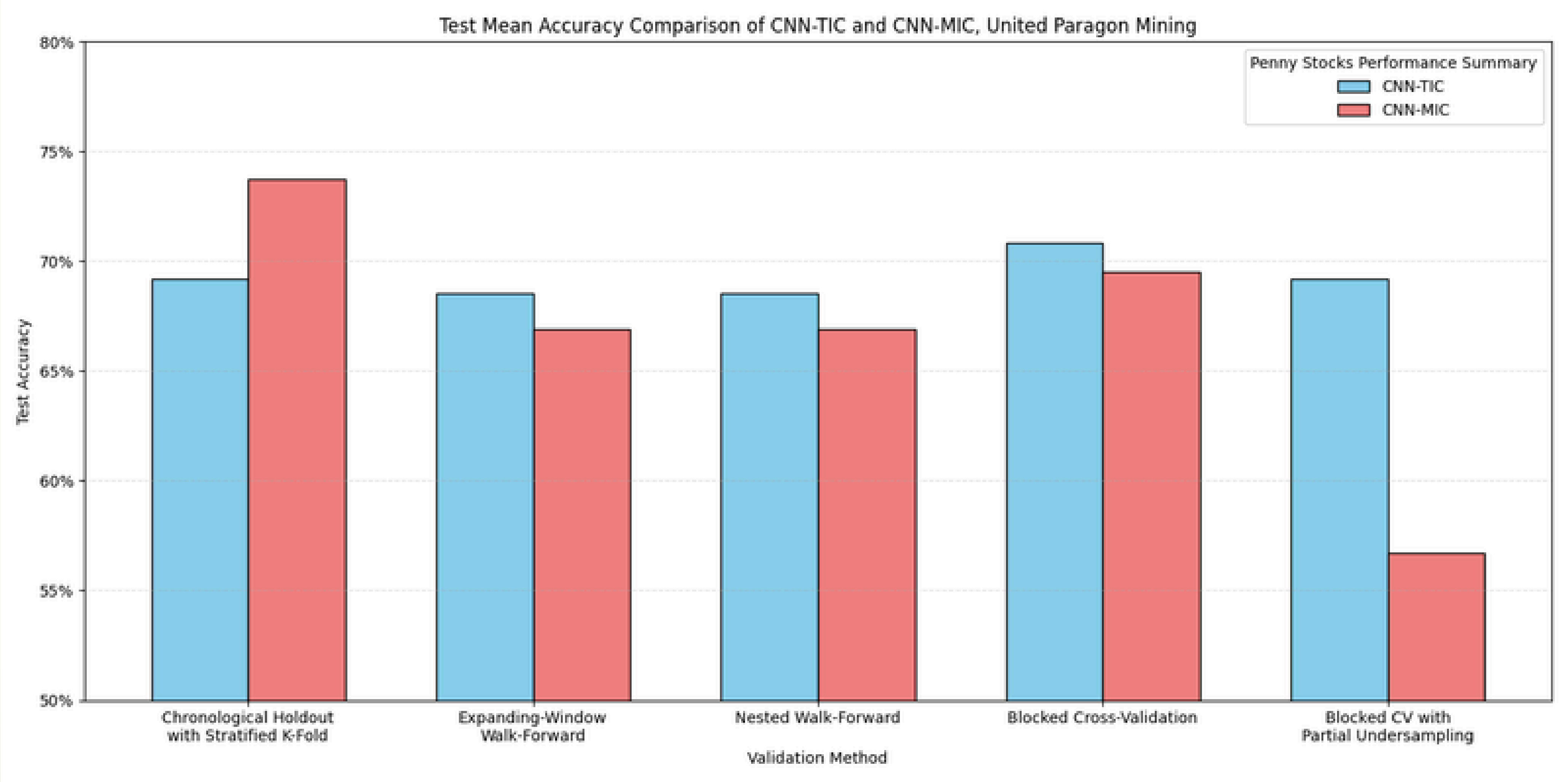
PH Resorts Group



Araneta Properties

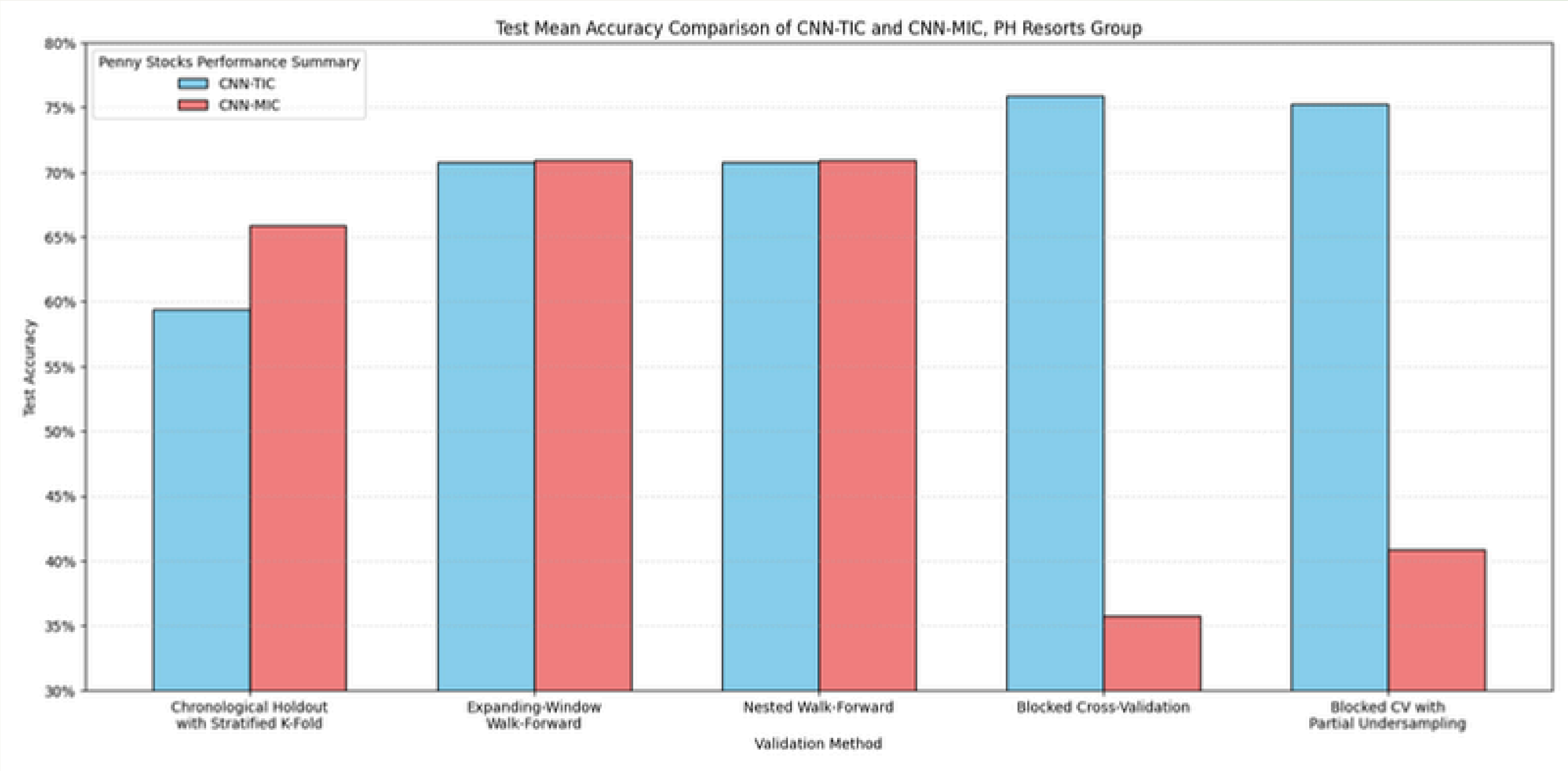


TEST ACCURACY

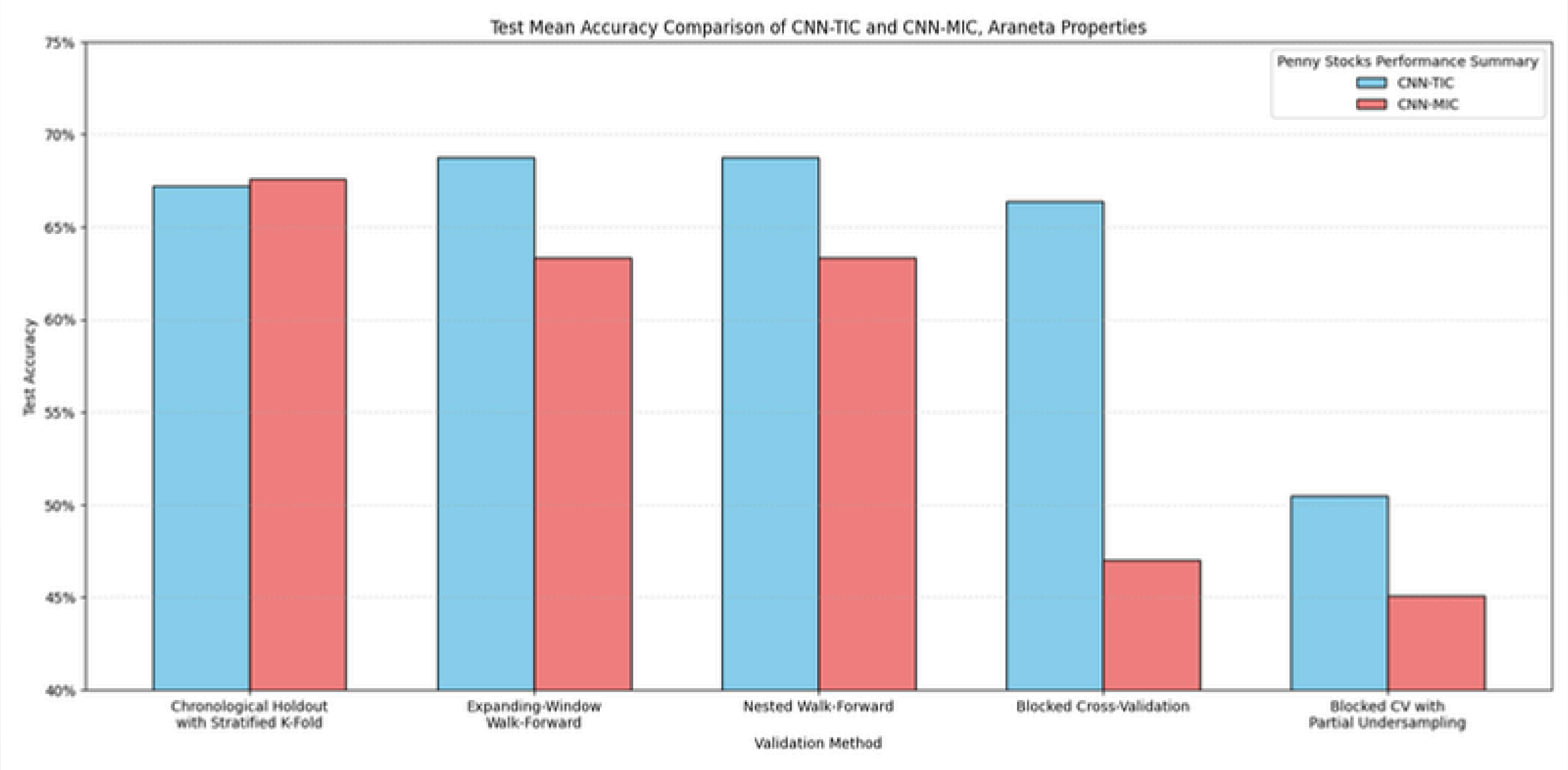


United Paragon Mining Corporation

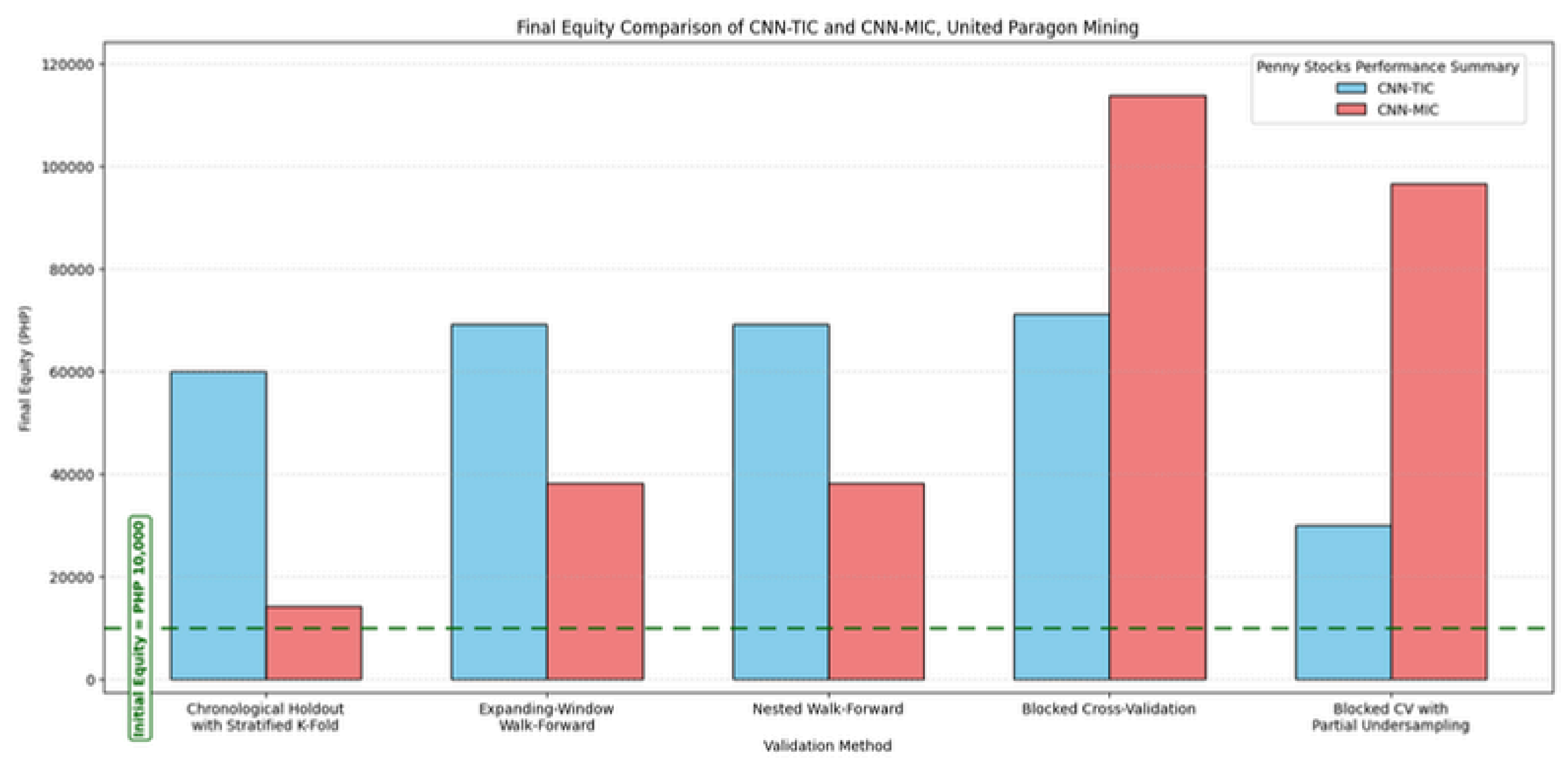
PH Resorts Group



Araneta Properties

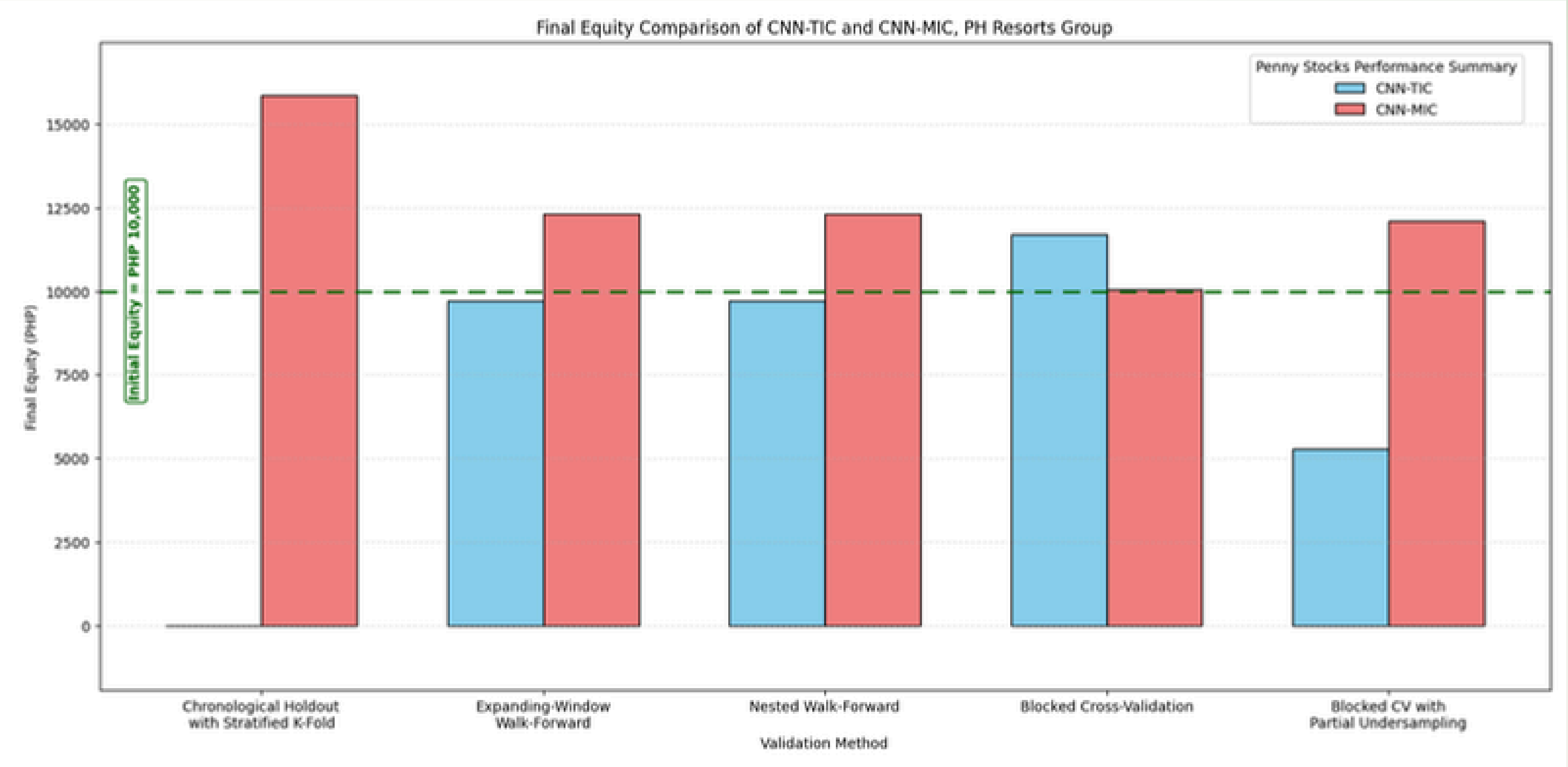


FINANCIAL PERFORMANCE

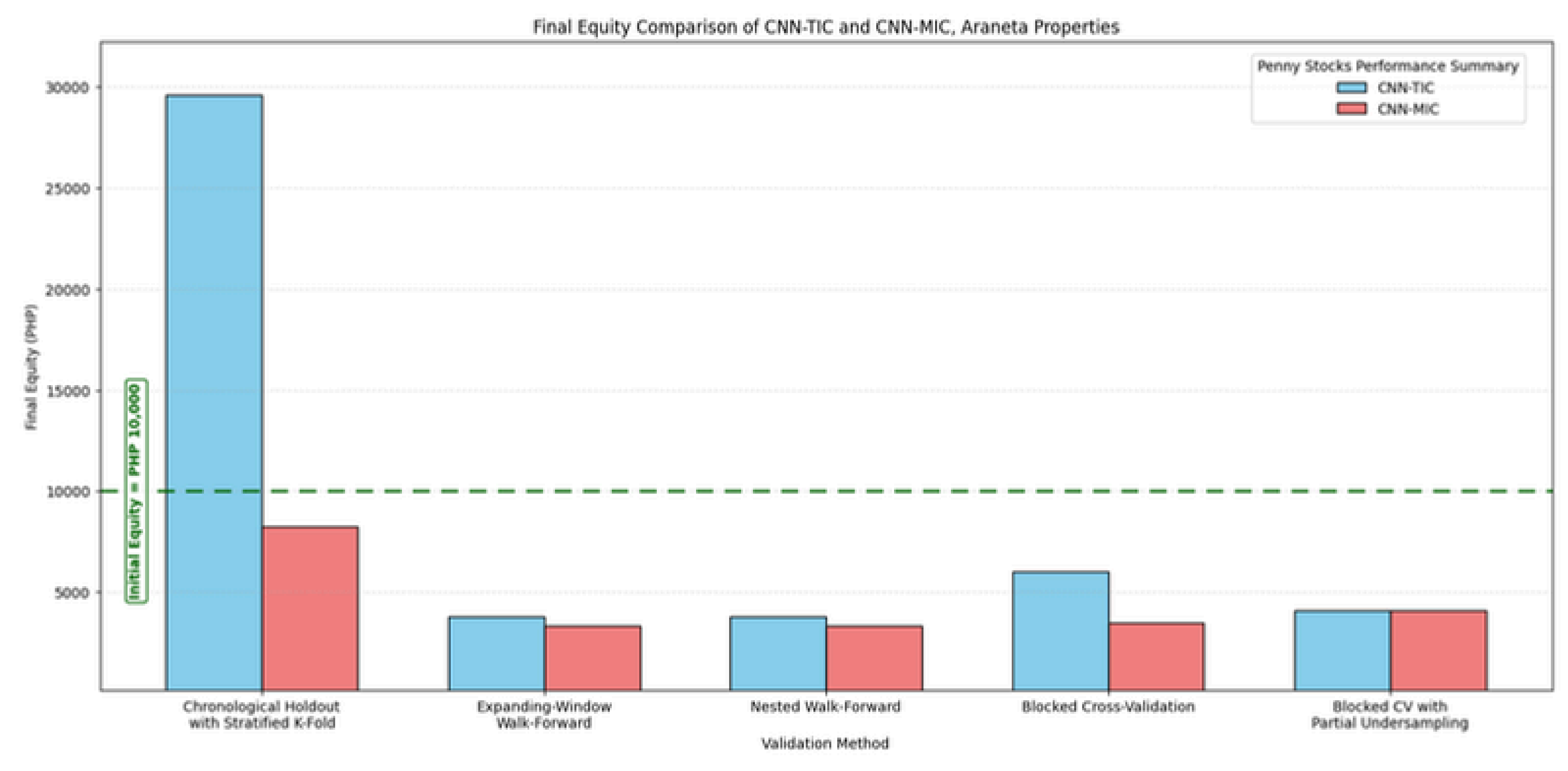


United Paragon Mining Corporation

PH Resorts Group



Araneta Properties

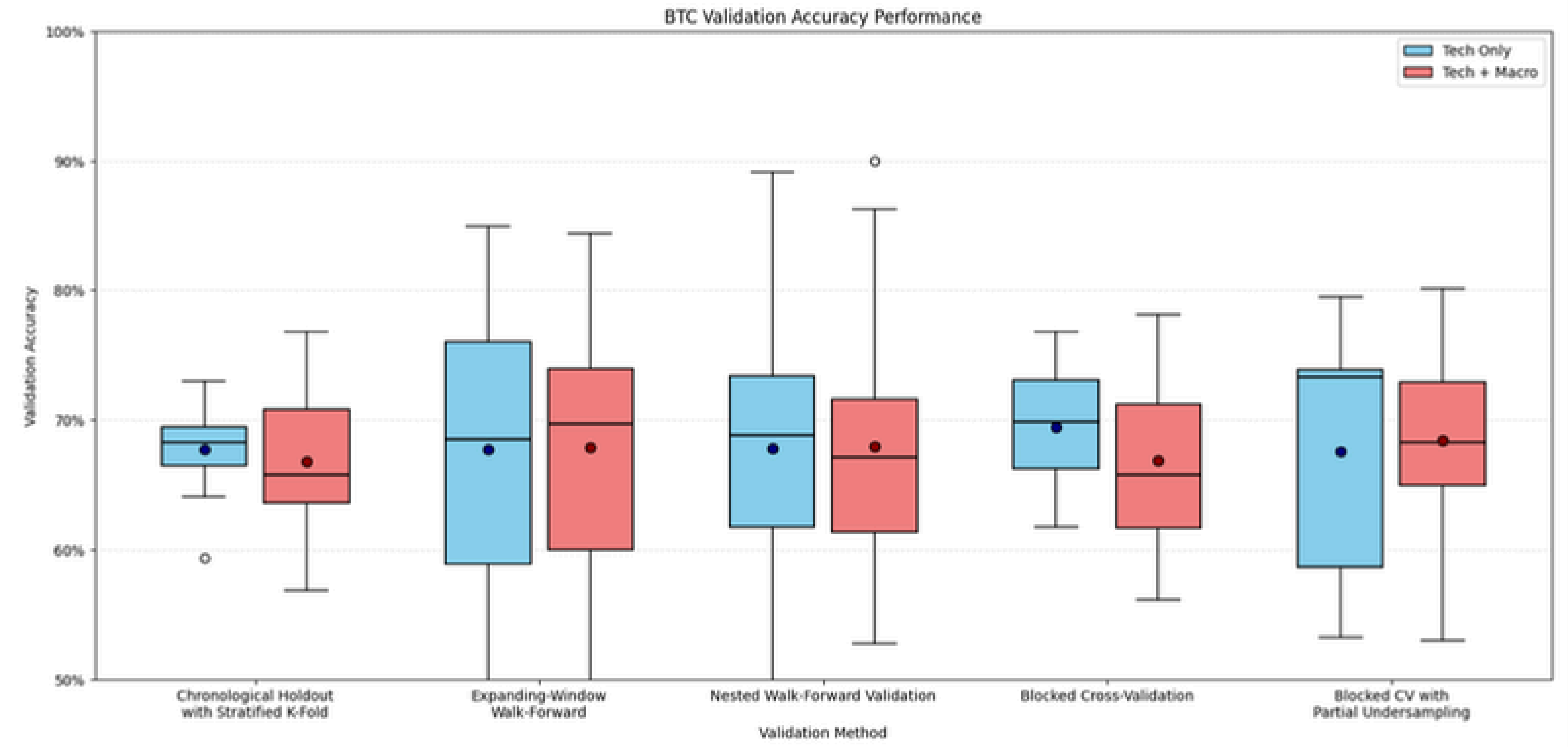


Cryptocurrency Training Results

Bitcoin

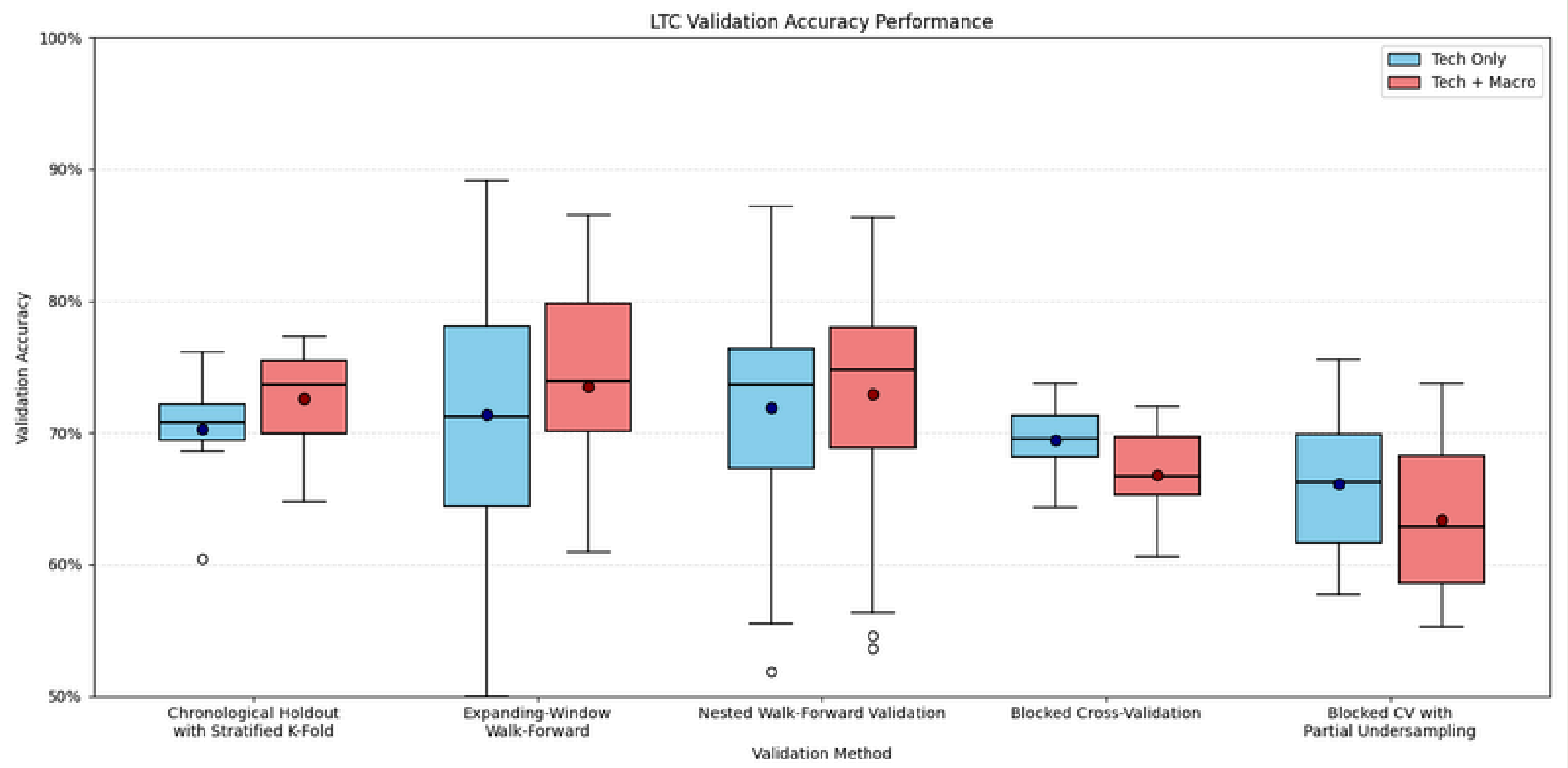
XRP

VALIDATION ACCURACY

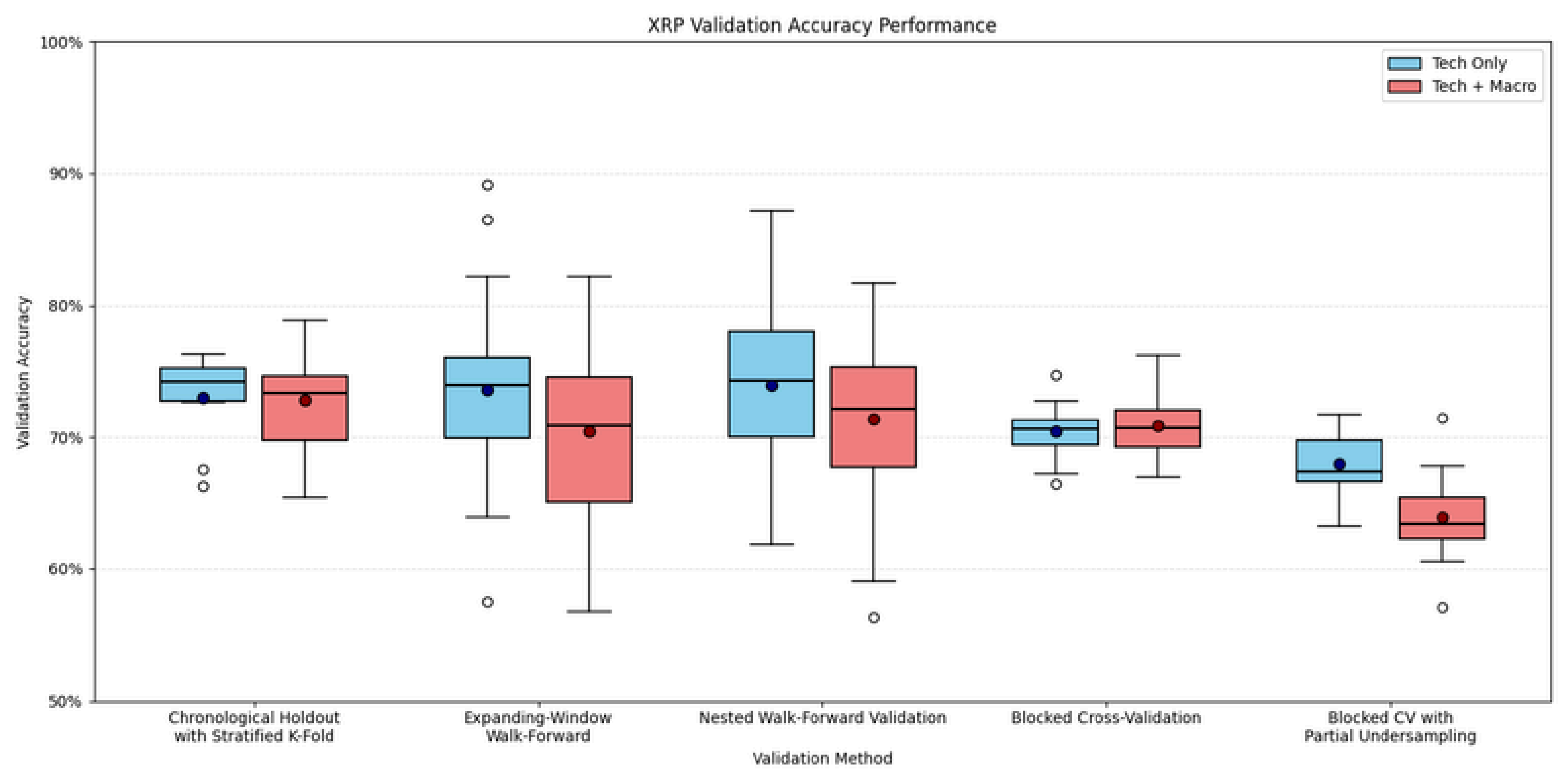


Bitcoin

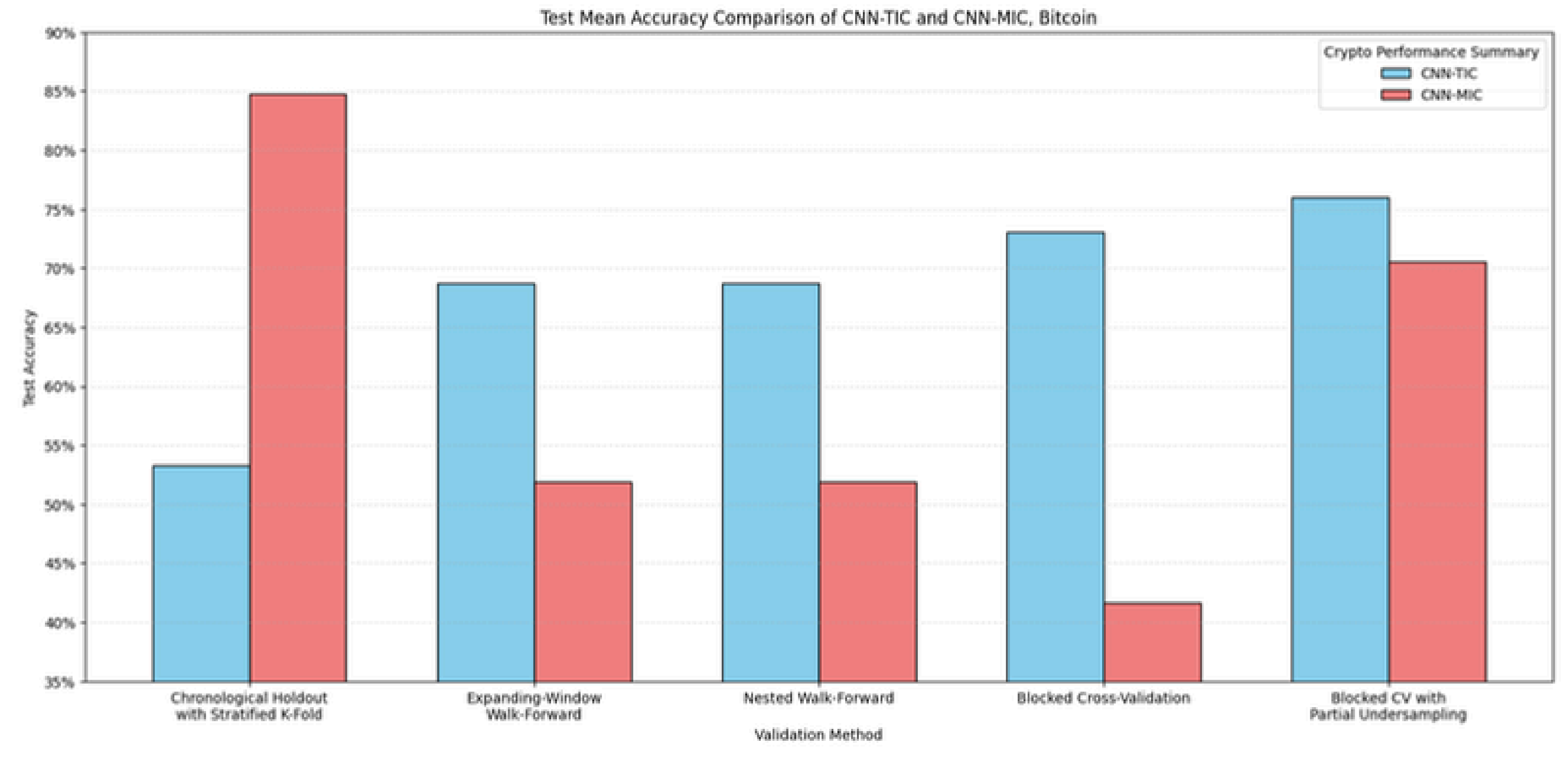
Litecoin



XRP

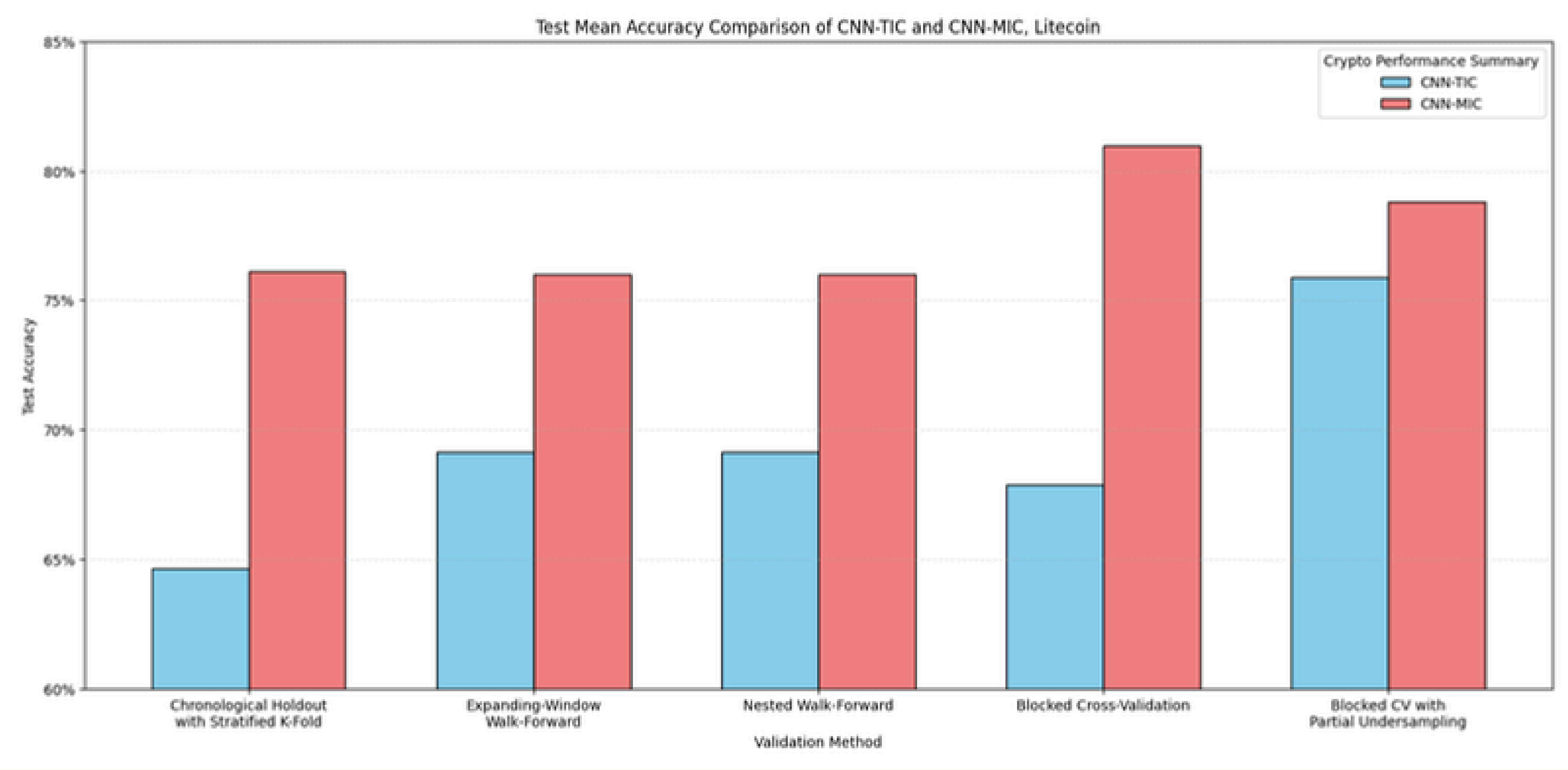

XRP Validation Accuracy Performance
Tech Only
Tech + Macro
100%
90%
80%
70%
60%
50%
Validation Accuracy
Chronological Holdout with Stratified K-Fold
Expanding-Window Walk-Forward
Nested Walk-Forward Validation
Blocked Cross-Validation
Blocked CV with Partial Undersampling
Validation Method

TEST ACCURACY

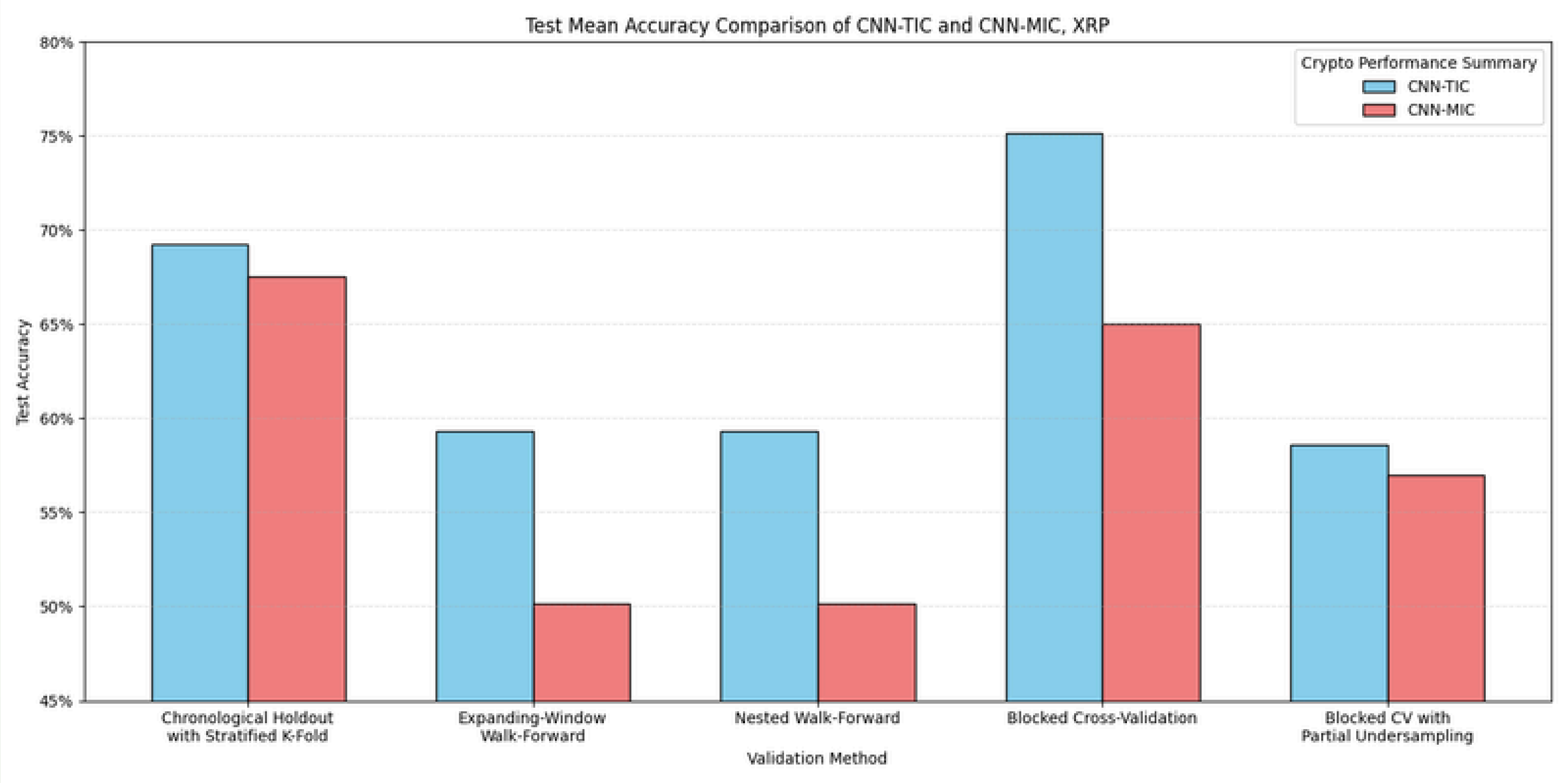


Bitcoin

Litecoin

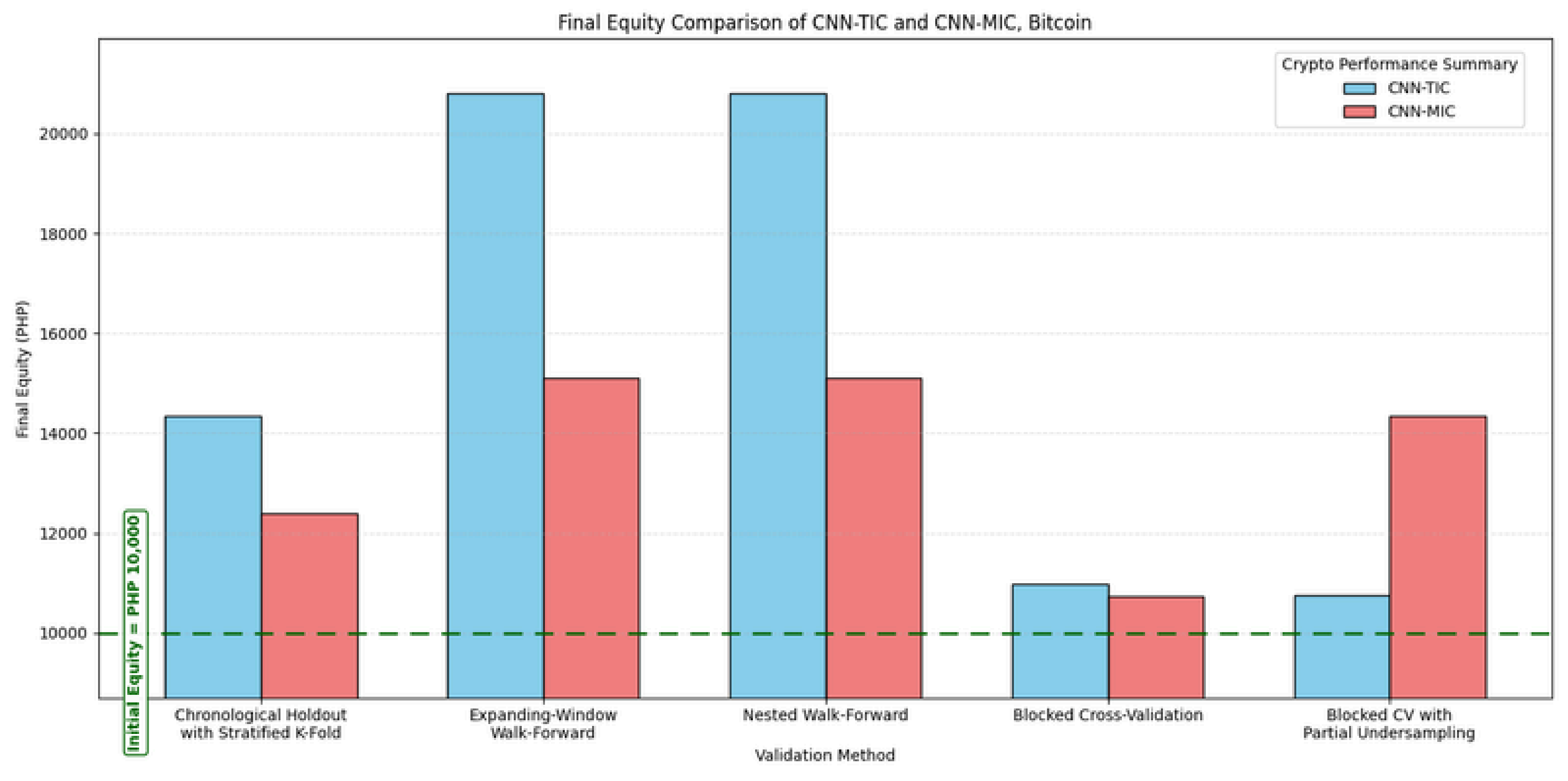


XRP

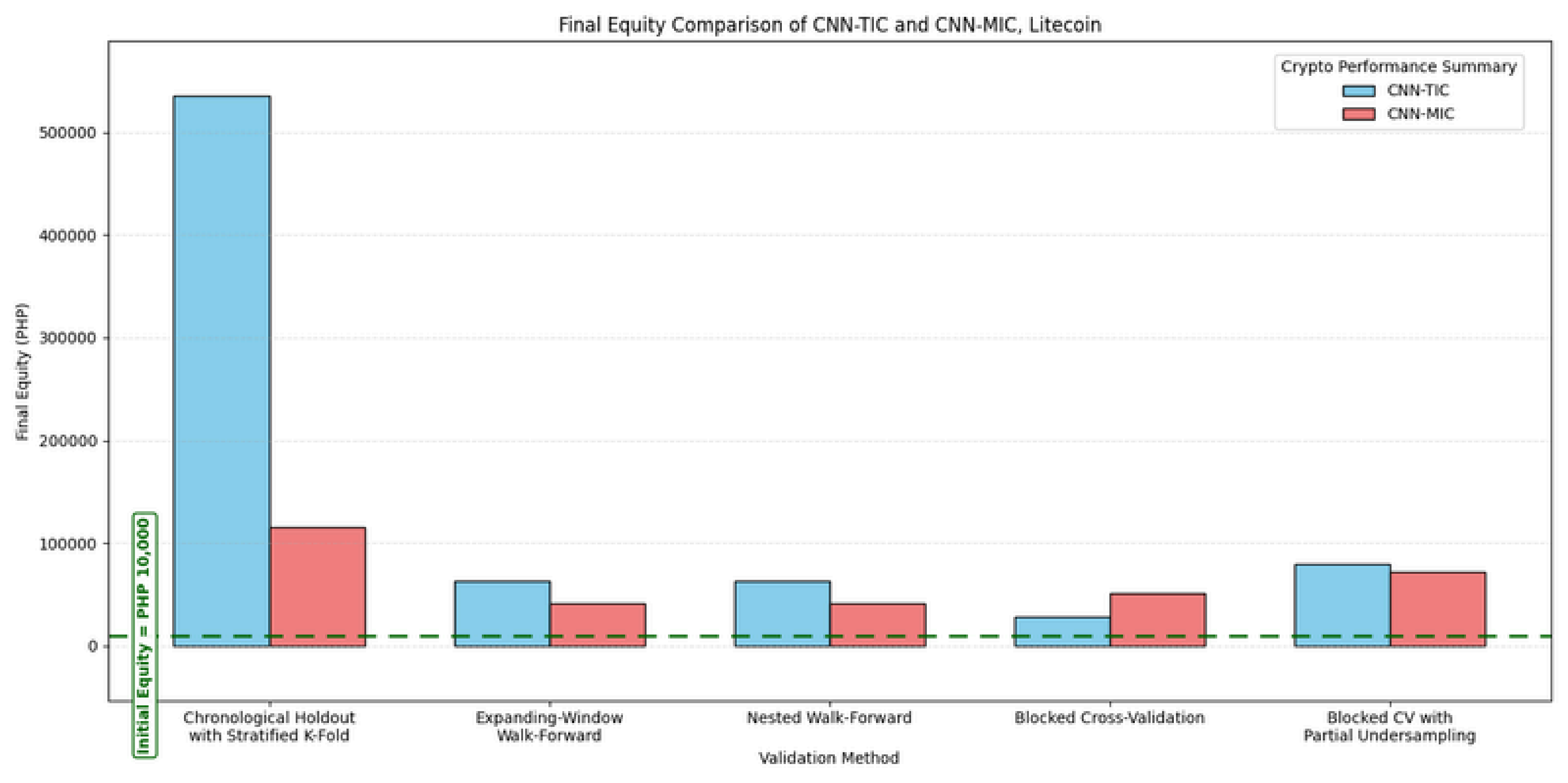

Test Mean Accuracy Comparison of CNN-TIC and CNN-MIC, XRP
Crypto Performance Summary
CNN-TIC
CNN-MIC
Test Accuracy
80%
75%
70%
65%
60%
55%
50%
45%
Chronological Holdout with Stratified K-Fold
Expanding-Window Walk-Forward
Nested Walk-Forward
Blocked Cross-Validation
Blocked CV with Partial Undersampling
Validation Method

FINANCIAL PERFORMANCE

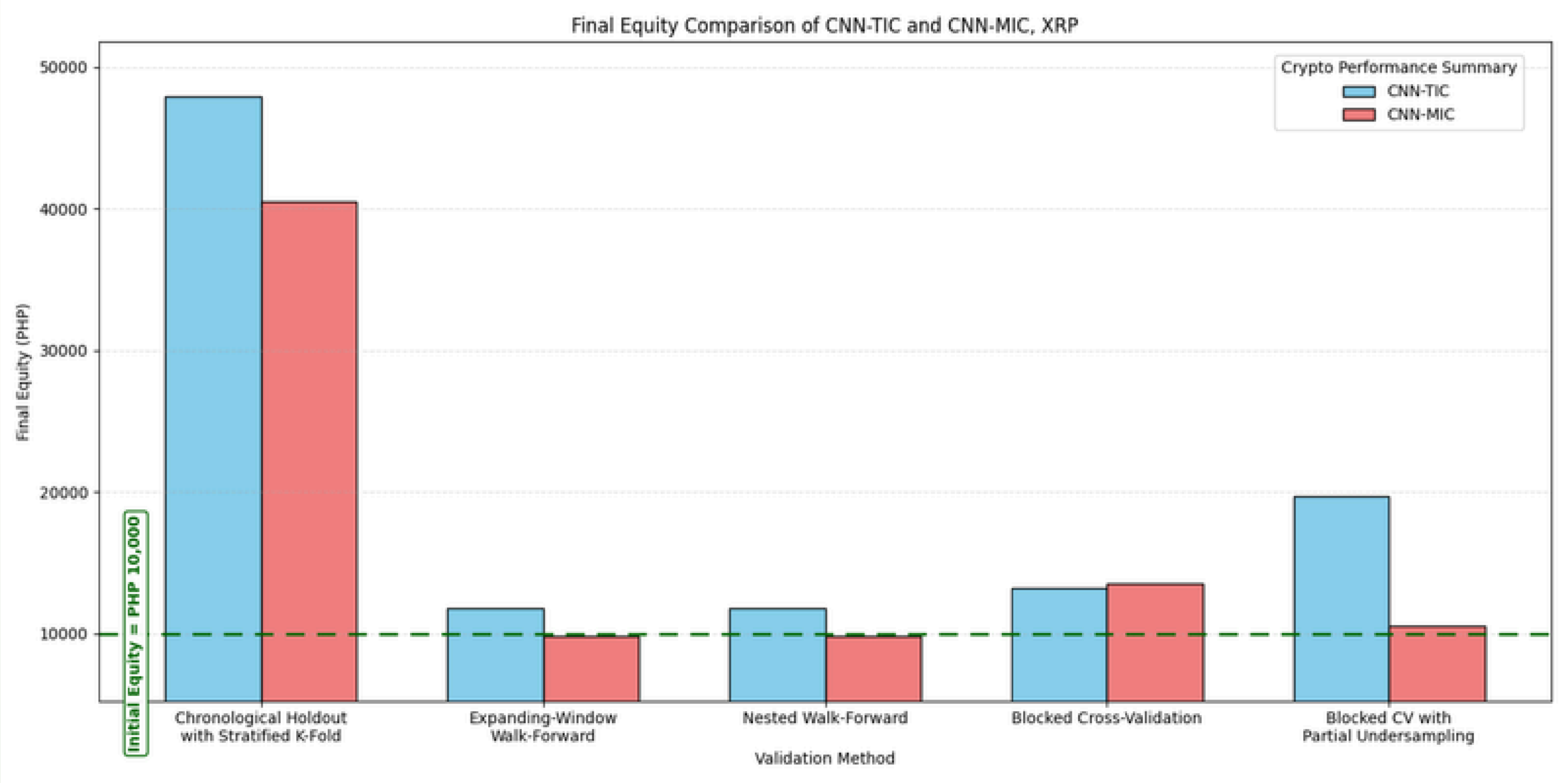
Bitcoin



Litecoin

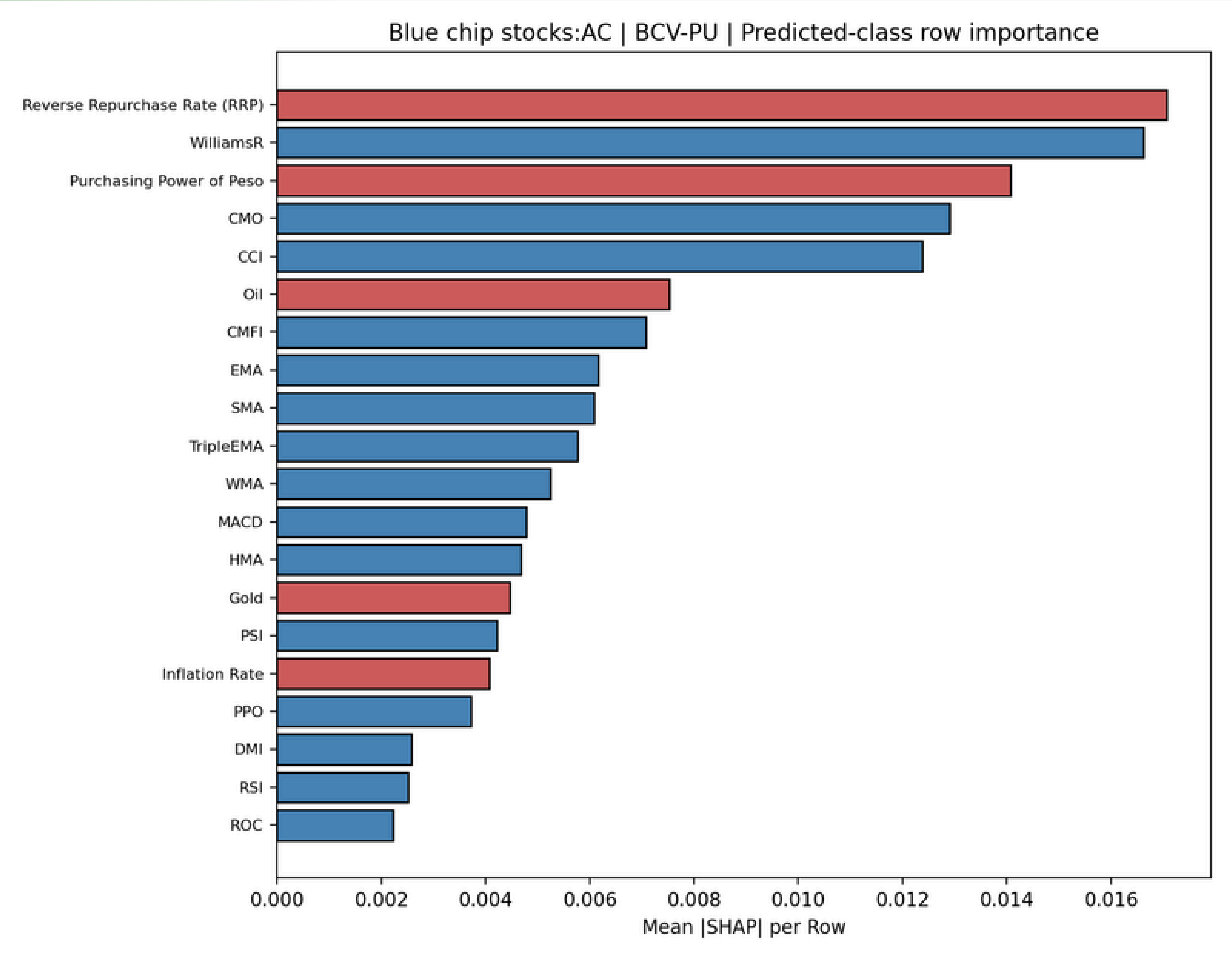


XRP


Final Equity Comparison of CNN-TIC and CNN-MIC, XRP
Crypto Performance Summary
CNN-TIC
CNN-MIC
Final Equity (PHP)
50000
40000
30000
20000
10000
Initial Equity = PHP 10,000
Chronological Holdout with Stratified K-Fold
Expanding-Window Walk-Forward
Nested Walk-Forward
Blocked Cross-Validation
Blocked CV with Partial Undersampling
Validation Method

SHAP ANALYSIS

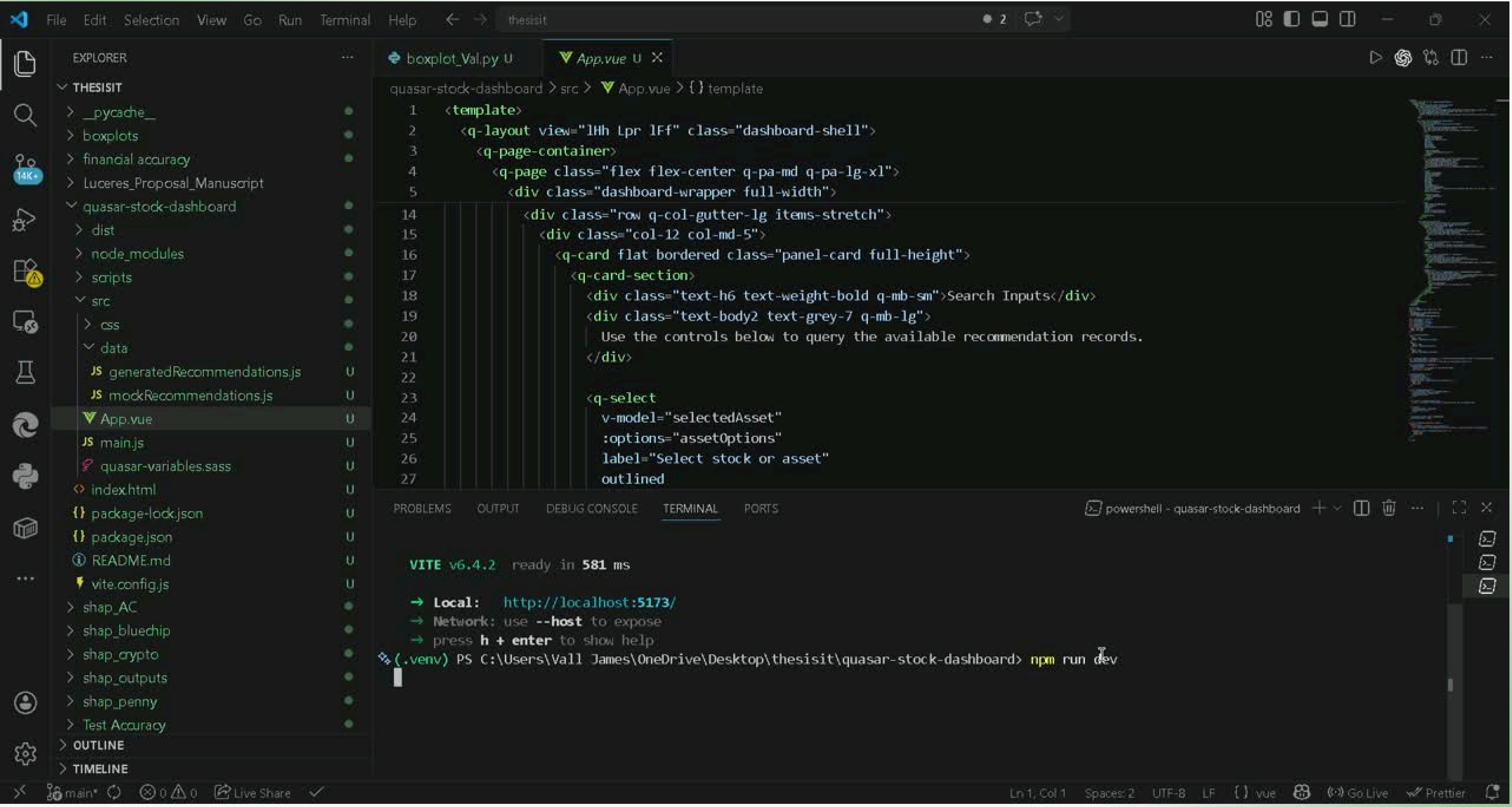
SHAP ANALYSIS



CONCLUSION

- The study proposed CNN-MIC, which extends CNN-TIC by integrating macroeconomic indicators into the same image representation.
- The results showed that macroeconomic indicator integration has potential value, but its benefit is not consistent across all assets and validation schemes.
- In selected cases, CNN-MIC produced competitive, and sometimes stronger classification and trading performance.
- The findings suggest that macroeconomic indicators can contribute information to CNN-based financial trend prediction.

Demo Video


File Edit Selection View Go Run Terminal Help
thesisit
EXPLORER
THESISIT
__pycache__
boxplots
financial accuracy
Luceres_Proposal_Manuscript
quasar-stock-dashboard
dist
node_modules
scripts
src
css
data
generatedRecommendations.js
mockRecommendations.js
App.vue
main.js
quasar-variables.sass
index.html
package-lock.json
package.json
README.md
vite.config.js
shap_AC
shap_bluechip
shap_crypto
shap_outputs
shap_penny
Test Accuracy
OUTLINE
TIMELINE
boxplot_Val.py U
App.vue U
quasar-stock-dashboard > src > App.vue > {} template
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<q-card flat bordered class="panel-card full-height">
<q-card-section>
<div class="text-h6 text-weight-bold q-mb-sm">Search Inputs</div>
<div class="text-body2 text-grey-7 q-mb-lg">
Use the controls below to query the available recommendation records.
</div>
<q-select
v-model="selectedAsset"
:options="assetOptions"
label="Select stock or asset"
outlined
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - quasar-stock-dashboard
VITE v6.4.2 ready in 581 ms
→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
(.venv) PS C:\Users\Vall James\OneDrive\Desktop\thesisit\quasar-stock-dashboard> npm run dev
main* Live Share
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References

- Chakole, J., & Kurhekar, M. P. (2021). Convolutional Neural Network-based a novel deep trend following strategy for stock market trading. CIKM Workshops.
- Cohen, N., Balch, T., & Veloso, M. (2020). Trading via image classification. In Proceedings of the First ACM International Conference on AI in Finance (pp. 1–6).
- Liu, J., & Kang, Y. (2023). Automated cryptocurrency trading approach using ensemble deep reinforcement learning: Learn to understand candlesticks. SSRN. https://doi.org/10.2139/ssrn.4348791
- Ma, F., et al. (2022). Macroeconomic attention and stock market return predictability. Journal of International Financial Markets, Institutions and Money, 79, 101603.
- Maratkhan, A., et al. (2021). Deep learning-based investment strategy: Technical indicator clustering and residual blocks. Soft Computing, 25(7), 5151–5161.
- Mo, F., Yan, S., & Xiao, Y. (2024). MoF: A background-aware multi-source fusion financial trend forecasting mechanism. Computational Economics, 1–30.
- Modak, R. (2023). Market sentiment analysis using multimodal transformers: Integrating earnings calls, social media, and technical indicators. Global Journal of Engineering and Technology Advances. https://api.semanticscholar.org/CorpusID:279909589
- Saud, A. S., & Shakya, S. (2024). Technical indicator empowered intelligent strategies to predict stock trading signals. Journal of Open Innovation: Technology, Market, and Complexity, 10(4), 100398.
- Sezer, O. B., & Ozbayoglu, A. M. (2018). Algorithmic financial trading with deep convolutional neural networks: Time series to image conversion approach. Applied Soft Computing, 70, 525–538.
- Ta, D. L. P., & contributors. (2025). Technical Analysis Library in Python (ta) (Version latest) [Python package]. GitHub. https://github.com/bukosabino/ta
- Wang, H., Lu, S., & Zhao, J. (2019). Aggregating multiple types of complex data in stock market prediction: A model-independent framework. Knowledge-Based Systems, 164, 193–204.
- Chauhan, S. S., et al. (2025). Exploring the relationship between macroeconomic indicators and sectoral indices of the Indian stock market. F1000Research, 14, 180.

Thank you!

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